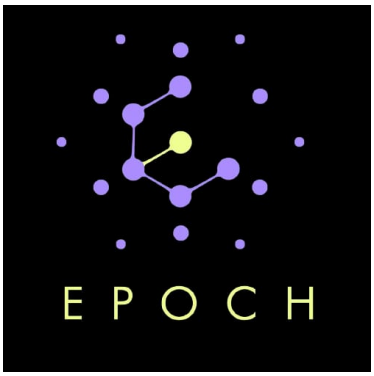
Math for ML Workshop Handouts and Exercises

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1 Introduction

Mathematics is an essential component of machine learning. Machine learning algorithms are based on mathematical models that enable them to make predictions, classifications, and other types of analyses. The following are some specific areas of mathematics that are crucial for machine learning:

1. **Probability theory**: Probability theory is used to model uncertainty in machine learning. It is used to calculate the likelihood of an event occurring, which is crucial for making predictions.

2. **Statistics**: Statistics is used to analyze data in machine learning. It includes techniques such as regression, hypothesis testing, and Bayesian inference.

3. **Linear algebra**: Linear algebra is used to represent and manipulate data in machine learning. It involves working with vectors, matrices, and linear transformations, which are used extensively in machine learning algorithms.

4. **Calculus**: Calculus is used in machine learning to optimize models and algorithms. It enables us to find the minimum or maximum value of a function, which is essential for training machine learning models.

In summary, mathematics plays a crucial role in machine learning, and a solid understanding of mathematical concepts is essential for building effective machine learning models.

2 Probability

2.1 Intro to Probability

Probability is used to model *uncertainty*. It indicates how likely an event is to occur or not to occur. A very elementary definition of probability that is typically taught in schools is

$$\text{Probability} = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

Probability has applications in several fields including (but not limited to) statistics, computer science, artificial intelligence, machine learning, data science, finance, quantum mechanics, game theory, marketing and social sciences.
This vast reach of probability makes it necessary for us to mathematically formalize the notion of probability. Without such formulation, it is unclear what the terms *event* and *outcome* mean.

2.1.1 Definitions

- A **random experiment** is a procedure that can be infinitely repeated and has a fixed, well-defined set of more than one possible outcomes
- Each repetition is known as a **trial**
- The above set of possible outcomes is known as the **sample space** of the experiment and is typically denoted by Ω
- An **event** is any subset of the sample space (can be the null set or even the complete set Ω itself)

$$A \subseteq \Omega$$

2.2 Elementary Set Theory

A set is simply put, a collection of distinct elements, which can be numbers or even other sets! Sets are fundamental to understanding probability. The following is a brief refresher of some of the important properties and operations of sets:

- **Membership:** $x \in A$ denotes that x is a member (element) of the set A
- **Cardinality:** $|A|$ denotes the number of elements in the set A
- **Union:** $A \cup B = \{x : x \in A \textbf{ or } x \in B\}$
- **Intersection:** $A \cap B = \{x : x \in A \textbf{ and } x \in B\}$
- **Cartesian product:** $A \times B = \{(x, y) : x \in A \textbf{ and } y \in B\}$

- **Empty set:** $\phi = \{\}$

$$A \cup \phi = A$$
$$A \cap \phi = \phi$$

- **Complement:** $A^c = \{x : x \notin A\}$

$$A \cup A^c = \Omega$$
$$A \cap A^c = \phi$$

- **Set difference:** $A \setminus B = \{x : x \in A \textbf{ and } x \notin B\} = A \cap B^c$
- **Symmetric difference:** Symmetric difference of the sets A and B is the set of elements that are in exactly one of A or B

$$\begin{aligned} A \,\Delta\, B &= \{x : (x \in A \textbf{ and } x \notin B) \textbf{ or } (x \in B \textbf{ and } x \notin A)\} \\ &= (A \setminus B) \cup (B \setminus A) \\ &= (A \cup B) \setminus (A \cap B) \end{aligned}$$

- **Subset:** $A \subseteq B$ if $x \in A \implies x \in B$
- **Proper subset:** $A \subset B$ if $A \subseteq B$ and $A \neq B$
- **Power set:** $\mathcal{P}(A) = \{X : X \subseteq A\}$
- **Inclusion-exclusion principle:** $|A \cup B| = |A| + |B| - |A \cap B|$
- **De Morgan's laws:**

$$(A \cup B)^c = A^c \cap B^c$$
$$(A \cap B)^c = A^c \cup B^c$$

- **Commutativity:**

$$A \cup B = B \cup A$$
$$A \cap B = B \cap A$$

- **Associativity:**

$$(A \cup B) \cup C = A \cup (B \cup C)$$
$$(A \cap B) \cap C = A \cap (B \cap C)$$

- **Distributivity:**

$$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$$
$$(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$$

2.3 Countability of Sets

A set A is said to be countable if and only if one of the following conditions is true:

- $|A|$ is finite
- $|A|$ is infinite but there is a bijective mapping $f : \mathbb{N} \to A$

The above mapping is essentially a way to **index** the set, which means that its elements can be counted in that particular order. This is the motivation behind defining countability - **defining an ordering on a set**.
From the definition of countability, we see that the sets $\{0, 1\}$ and $\mathbb{N}$ are countable trivially, the former being finite and for the latter, the identity mapping $f(x) = x$ exists.
Now consider the set of integers $\mathbb{Z}$. This set is also countable because of the following bijective mapping $f : \mathbb{N} \to \mathbb{Z}$

$$f(x) = \begin{cases} \frac{x-1}{2} & x \text{ is odd} \\ -\frac{x}{2} & x \text{ is even} \end{cases} \tag{1}$$

This mapping essentially *counts* the set of integers in the following *order*:

$$0, -1, 1, -2, 2, \ldots$$

The set of rational numbers $\mathbb{Q}$ is also countable. Observe that every positive rational number will be present in the following table because every rational number is of the form $\frac{a}{b}$ for some integers a and b. The ones that appear multiple times (where $\gcd(a, b) \neq 1$) can simply be crossed out. This will provide us with an ordering for *every* positive rational number. This argument can be extended to negative rational numbers just like what we did for negative integers.
We have seen some examples of countably infinite sets. Let us look at an uncountable set now - the set of real numbers in $(0, 1)$. In order to prove that $(0, 1)$ is uncountable, we will use the basic motivation behind countability - we will prove that we cannot order this set. Assume for the sake of contradiction that such an ordering exists, which means we can make a list of *every* real number in $(0, 1)$. Suppose the list of the numbers (in decimal notation) looks like the following:

$$\begin{aligned} a_1 &= 0.a_{11}a_{12}a_{13}\ldots \\ a_2 &= 0.a_{21}a_{22}a_{23}\ldots \\ a_3 &= 0.a_{31}a_{32}a_{33}\ldots \\ &\vdots \end{aligned}$$

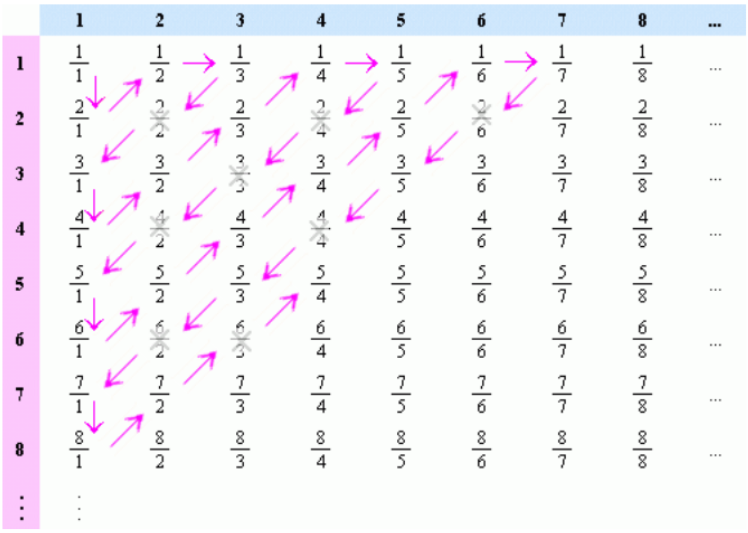


Figure 1: Ordering of the rationals

Now consider another real number $b \in (0, 1)$ whose digits are defined as follows:

$$b_i = \begin{cases} 2 & a_{ii} = 1 \\ 1 & a_{ii} \neq 1 \end{cases}$$

$$\text{i.e., } b_i \neq a_{ii} \qquad \forall i \in \mathbb{N}$$

Observe that $b \neq a_1$ because their first digits differ. Similarly, $b \neq a_2$ because their second digits differ. It is easy to see that $b \neq a_i \ \forall i \in \mathbb{N}$ which implies that b is not present in the above list. This is a contradiction because the list was supposed to contain every real number in $(0, 1)$ and hence, $(0, 1)$ is uncountable. This is called as the **Cantor's diagonal argument**.

2.3.1 Cardinality of infinite sets

Since infinite sets have infinitely many elements, their cardinality is simply infinity. However, *all infinities are not the same.*

Two sets A and B are said to have the same cardinality, i.e., $|A| = |B|$ if and only if there exists a bijection between them. Thus, for every countably infinite set A, we have $|A| = |\mathbb{N}|$. $|N|$ is usually denoted by $\aleph_0$ (aleph-null). Any uncountable set has a cardinality greater than $\aleph_0$.

Schröder-Bernstein Theorem: If there are injections $f : A \to B$ and $g : B \to A$ then there exists a bijection between A and B, i.e., $|A| = |B|$

2.3.2 Subset Theorem

Theorem: The subset of a countable set is countable
Proof: Consider a countable set A and another set $B \subseteq A$
If $|A|$ is finite then $|B| \leq |A|$ will be finite too which would make it countable
If $|A|$ is infinite and $|B|$ is finite, then B is still countable
Otherwie, if $|A|$ and $|B|$ are both infinite, then we have a bijection $f : \mathbb{N} \to A$
Since A is countable, we can list out the elements of A in the order

$$f(1), f(2), f(3), \ldots$$

Such a list is known as an **enumeration** of A
And since $B \subseteq A$, every element of B will appear in the above list
Define another mapping $g : \mathbb{N} \to B$ such that $g(i)$ is the ith element of B that appeared in the list
g is an injection because the elements of A are distinct
g is a surjection because *every* element of B will appear in the list at some point
Hence, $g : \mathbb{N} \to B$ is a bijection and B is countable

Corollary: The superset of an uncountable set is uncountable
Proof: This statement is the contrapositive of the above theorem
Consider $B \subseteq A$
We have proved that A is countable $\implies$ B is countable
Truthness of a statement implies the truthness of its contrapositive
Therefore, B is uncountable $\implies$ A is uncountable

Note that from this theorem, it follows that the set of real numbers $\mathbb{R}$ is uncountable because $\mathbb{R} \supseteq (0, 1)$

2.3.3 Union Theorem

Lemma: The union of two countable sets is countable
Proof: Consider countable sets A and B
If they are both finite, then $A \cup B$ is also finite and thus, countable
If only one of them is finite (say A) then we can define the following ordering on $A \cup B$ where $m = |A|$

$$a_1, a_2, \ldots, a_m, b_1, b_2, b_3, \ldots$$

i.e., we get the following bijection $f : \mathbb{N} \to A \cup B$

$$f(i) = \begin{cases} a_i & 1 \leq i \leq m \\ b_{i-m} & i > m \end{cases}$$

If both of them are infinite, then the following ordering can be defined

$$a_1, b_1, a_2, b_2, \ldots$$

i.e., we get the following bijection $f : \mathbb{N} \to A \cup B$

$$f(i) = \begin{cases} a_{\frac{i+1}{2}} & i \text{ is odd} \\ b_{\frac{i}{2}} & i \text{ is even} \end{cases}$$

Theorem: The union of finitely many countable sets is countable
Proof: This follows via strong induction
Consider n countable sets $A_1, \ldots, A_n$
From the previous lemma, $A_1 \cup A_2$ is countable
Since A_3 is countable, $(A_1 \cup A_2) \cup A_3$ is countable and so on

2.4 Axioms of Probability

The three fundamental axioms of probability, also known as **Kolmogorov axioms** can be used to derive every property related to probability.
Let Ω be the sample space and $A \subseteq \Omega$ be any event

1. $\Pr(A) \geq 0 \quad \forall\ A \subseteq \Omega$
2. $\Pr(\Omega) = 1$
3. If $A_1, A_2, \ldots$ is a countable sequence of disjoint events, i.e., $A_i \cap A_j = \phi\ \forall i, j$ then

$$\Pr\left(\bigcup_i A_i\right) = \sum_i \Pr(A_i)$$

2.5 Probability Properties

The following are some of the most important properties of probability, all of which can be derived from the axioms.

1. If $A \subseteq B$ then $\Pr(A) \leq \Pr(B)$

   Proof: $A \subseteq B \implies B = A \cup (B \setminus A)$
   Since A and $B \setminus A$ are disjoint, $\Pr(B) = \Pr(A) + \Pr(B \setminus A)$ by Axiom 3
   Also $\Pr(B \setminus A) \geq 0$ by Axiom 1
   Thus, $\Pr(B) \geq \Pr(A)$

2. $\Pr(\phi) = 0$

   Proof: Consider any event $A \subseteq \Omega$
   Then $A = A \cup \phi$
   Since $A \cap \phi = \phi$, they are disjoint
   Thus, $\Pr(A) = \Pr(A) + \Pr(\phi) \implies \Pr(\phi) = 0$

3. $\Pr(A^c) = 1 - \Pr(A)$

   Proof: $A \cup A^c = \Omega \implies \Pr(A) + \Pr(A^c) = \Pr(\Omega)$ by Axiom 3 since A and A^c are disjoint
   $\Pr(\Omega) = 1$ by Axiom 2
   Thus, $\Pr(A) + \Pr(A^c) = 1$

4. $0 \leq \Pr(A) \leq 1 \quad \forall\ A \subseteq \Omega$

   Proof: $\Pr(A) \geq 0$ by Axiom 1
   $\Pr(A) = 1 - \Pr(A^c)$ But $\Pr(A^c) \geq 0$ by Axiom 1
   Thus, $\Pr(A) \leq 1$

5. $\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$

   Proof: $A \cup B = (A \setminus B) \cup (B \setminus A) \cup (A \cap B)$
   These three sets are disjoint
   By Axiom 3, $\Pr(A \cup B) = \Pr(A \setminus B) + \Pr(B \setminus A) + \Pr(A \cap B)$
   Also $\Pr(A \setminus B) + \Pr(A \cap B) = \Pr(A)$
   and $\Pr(B \setminus A) + \Pr(A \cap B) = \Pr(B)$
   Thus,

$$\begin{aligned}\Pr(A \cup B) &= (\Pr(A \setminus B) + \Pr(A \cap B)) + (\Pr(B \setminus A) + \Pr(A \cap B)) - \Pr(A \cap B) \\ &= \Pr(A) + \Pr(B) - \Pr(A \cap B)\end{aligned}$$

2.6 Conditional Probability

Conditional probability is the basis of Bayesian learning. The conditional probability of an event A given B where $A, B \subseteq \Omega, \Pr(B) \neq 0$ is defined as

$$\Pr(A \mid B) \triangleq \frac{\Pr(A \cap B)}{\Pr(B)}$$

2.7 Independence of Events

Two events A and B where $A, B \in \Omega$ are said to be independent if and only if

$$\Pr(A \cap B) = \Pr(A) \Pr(B)$$

In such a case

$$\Pr(A \mid B) = \frac{\Pr(A) \Pr(B)}{\Pr(B)} = \Pr(A)$$

An event A is independent of itself if it is either certain or impossible

$$\begin{aligned}&\Pr(A \cap A) = \Pr(A) \Pr(A) \\ &\text{i.e., } \Pr(A) = \Pr(A)^2 \\ &\text{i.e., } \Pr(A) = 0 \text{ or } 1\end{aligned}$$

A countable sequence of events $A_1, A_2, \ldots$ is said to be pairwise independent if and only if

$$\Pr(A_i \cap A_j) = \Pr(A_i) \Pr(A_j) \quad \forall\ i, j$$

A countable sequence of events $A_1, A_2, \ldots$ is said to be mutually independent if and only if

$$\Pr\left(\bigcap_i A_i\right) = \prod_i \Pr(A_i)$$

Mutual independence implies pairwise independence but the converse is not true

2.8 Random Variables

To formalize the notion of an outcome of the trial, we assign a number to every possible outcome. A **random variable** is a mapping (function) from the set of possible outcomes, i.e., the sample space, to the reals

$$X : \Omega \to \mathbb{R}$$

Random variables are typically represented by uppercase letters and their instantiations are represented by lowercase letters, e.g., for some $\omega \in \Omega$

$$X(\omega) = x$$

Probabilities are associated with events. For example, consider the events $\{\omega : X(\omega) = 7\}$ and $\{\omega : X(\omega) \in \mathbb{N}\}$. The probabilities associated with these events are denoted by

$$\Pr(\{\omega : X(\omega) = 7\}), \qquad \Pr(\{\omega : X(\omega) \in \mathbb{N}\})$$

But in most places, you will find that probabilities are represented in a much simpler format, like the following

$$\Pr(X = 7), \qquad \Pr(X \in \mathbb{N})$$

However, it is important to remember that this is just an abuse of notation. Always keep in mind that events are sets and random variables are functions.

2.9 Random Vectors

As the name suggests, a **random vector** is a collection of several random variables defined on the **same sample space**.

$$\mathbf{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix}$$

where $X_i : \Omega \to \mathbb{R}$ for each $i = 1, \ldots, n$.

They are also known as **multivariate random variables**, which are typically represented by boldface uppercase letters. From ML perspective, it is often useful to analyze data in multivariate form where each X_i corresponds to a different *feature* of the data.

2.10 Types of Random Variables

1. **Discrete Random Variable:** A random variable whose **support** is *countable*.
   e.g., number of heads in a series of coin tosses, number of people outside mess on a Sunday night, etc

2. **Continuous Random Variable:** A random variable whose **support** is *uncountable*.
   e.g., amount of rainfall in Sangareddy, calories burnt after gym, etc

2.11 Probability Mass Function

Probability Mass Function (**PMF**) is the probability distribution of a discrete random variable (X) and used to associate probabilities for each real value that X can take.
Mathematically it is defined as follows,

$$p_X : \mathbb{R} \to [0, 1]$$
$$p_X(x) = \Pr(X = x)$$

Note : Here x may not lie in **support** of X in which case the $\Pr(X = x) = 0$.

2.11.1 Properties of PMF

- Naturally, a probability mass function is always non-negative

$$p_X(x) \geq 0 \quad \forall\, x \in \mathbb{R}$$

- The name *mass* comes from mass in Physics, just like physical mass, probability mass (the total probability) is conserved which here means that all the probability masses sum up to one

$$\sum_{x \in \mathbb{R}} p_X(x) = 1$$

  The PMF of a discrete random variable look like Fig : 2

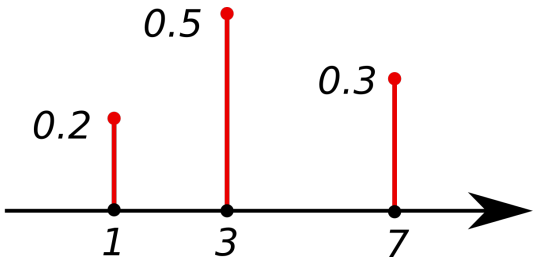


Figure 2: PMF of a discrete random variable

2.12 Cummulative Density Function in discrete case

The **cumulative distribution function** (CDF) of a random variable (X) represents the cumulative probability until a particular value. It is defined for both discrete and continuous random variables.
Mathematically, for discrete random variables, it is defined as

$$\begin{aligned} F_X &: \mathbb{R} \to [0,1] \\ F_X(x) &\stackrel{\triangle}{=} \Pr(X \leq x) \qquad (2) \\ &= \sum_{i \leq x} \Pr(X = i) \\ &= \sum_{i \leq x} p_X(i) \end{aligned}$$

where $p_X(x)$ is the PMF of random variable X
One can relate the probability of X lie in $(a, b]$ using the Cummulative Probability Function $F_X(x)$,

$$\Pr(a < X \leq b) = F_X(b) - F_X(a)$$

2.12.1 Properties of Cummulative Distribution Function

- Every Cummulative Distributive Function, $F_X(x)$ is a non-decreasing function

  $$x_1 < x_2 \implies F_X(x_1) \leq F_X(x_2)$$

  Proof: Consider,

  $$\begin{aligned} H_X(x_1, x_2) &= F_X(x_1) - F_X(x_2) \qquad (3) \\ &= \Pr(X \leq x_1) - \Pr(X \leq x_2) \qquad (4) \end{aligned}$$

  Given $x_1 < x_2$,

  $$\begin{aligned} &= \Pr(X \leq x_1) - (\Pr(X \leq x_1) + \Pr(x_1 < X \leq x_2)) \qquad (5) \\ &= -\Pr(x_1 < X \leq x_2) \leq 0 \qquad (6) \end{aligned}$$

  $\therefore$ When $x_1 < x_2 \implies F_X(x_1) \leq F_X(x_2)$

- Every Cummulative Distributive Function, $F_X(x)$ is right-continuous

$$F_X(x_0) = \lim_{x \to x_0^+} F(x)$$

Proof: Consider the limit,

$$\begin{aligned}
\lim_{x \to x_0^+} F_X(x) &= \lim_{x \to x_0^+} \Pr(X \leq x) && (7)\\
&= \lim_{h \to 0} \Pr(X \leq x_0 + h) && (8)\\
&= \lim_{h \to 0} (\Pr(X \leq x_0) + \Pr(x_0 < X \leq x_0 + h)) && (9)\\
&= \lim_{h \to 0} \Pr(X \leq x_0) + 0 && (10)\\
&= F_X(x) && (11)
\end{aligned}$$

In this way, we can prove that CDF is right continuous. One can prove that CDF may not be left continuous since it limit can only result to $\Pr(X < x)$.

- $$\lim_{x \to -\infty} F(x) = 0$$

Proof: It is trivial and follows from the definition

- $$\lim_{x \to \infty} F(x) = 1$$

Proof: It is trivial and follows from the definition

2.13 Expectation of a Random Variable

In probability theory, expected value is the generalization of weighted average. In the context of a random variable, **expected value** (a.k.a. mean μ) is the average value that one will observe after conducting many **independent** experiments.

2.13.1 Properties of Expectation

- If $X(w) \geq 0 \ \forall \ w \in \Omega$ then $\mathbb{E}[X] \geq 0$
- **Linearity of expectation**

$$\mathbb{E}[aX + bY] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y] \quad \forall \ a, b \in \mathbb{R}$$

where X and Y are random variables defined on the same sample space

- If $X(w) \geq Y(w) \ \forall \ w \in \Omega$ then $\mathbb{E}[X] \geq \mathbb{E}[Y]$
- $\mathbb{E}[|X|] \geq |\mathbb{E}[X]|$

2.13.2 Expectation of a Discrete Random Variable

Mathematically, the expectation of a discrete random variable is defined as

$$\mathbb{E}[X] \triangleq \sum_{x \in \mathbb{R}} x \cdot p_X(x)$$

where $p_X(x)$ is the PMF of the discrete random variable X
Similarly, the expectation of a function of a random variable $g(X)$ is defined as

$$\mathbb{E}[g(X)] \triangleq \sum_{x \in \mathbb{R}} g(x) \cdot p_X(x)$$

2.13.3 Expectation of a Random Vector

Suppose we have the random vector $\mathbf{X}$,

$$\mathbf{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix}$$

Then expectation of random vector $\mathbf{X}$ is defined as

$$\mathbb{E}[\mathbf{X}] \triangleq \begin{pmatrix} \mathbb{E}[X_1] \\ \vdots \\ \mathbb{E}[X_n] \end{pmatrix}$$

2.14 Variance of a Random Variable

Variance is a measure of **dispersion**, meaning it is a measure of how far a set of numbers is spread out from their **average** value. In the context of a random variable, variance measures the expected square of deviation from the mean of the population.
Mathematically it is defined as,

$$\operatorname{Var}[X] \triangleq \mathbb{E}\left[(X - \mathbb{E}[X])^2\right]$$

The square root of variance is known as **standard deviation** or **standard error**

$$\sigma \triangleq \sqrt{\operatorname{Var}[X]}$$

Note : Variance can be considered as expectation of function of a random variable X.

2.14.1 Properties of Variance

- $\text{Var}[X] \geq 0$ for any random variable X
- $\text{Var}[a] = 0$ where $a \in \mathbb{R}$ is some constant
- $\text{Var}[X + a] = \text{Var}[X]$
- $\text{Var}[aX] = a^2 \text{Var}[X]$
- Note that $\text{Var}[X + Y] \neq \text{Var}[X] + \text{Var}[Y]$ in general. If X and Y are *independent*, then the assertion is true
- $\text{Var}[X] = \mathbb{E}\left[X^2\right] - (\mathbb{E}[X])^2$
  Proof: Consider the variance of X,

$$\text{Var}[X] = \mathbb{E}\left[X^2 + \mathbb{E}[X]^2 - 2X\mathbb{E}[X]\right]$$

  Using the linearity of expectation,

$$\begin{aligned}\text{Var}[X] &= \mathbb{E}\left[X^2\right] + \mathbb{E}[X]^2 - 2\mathbb{E}[X]^2 \\ &= \mathbb{E}\left[X^2\right] - \mathbb{E}[X]^2\end{aligned}$$

2.15 Examples of Discrete Random Variable

1. **<u>Bernoulli Random Variable :</u>**
   - Discrete Random Variable (X) whose support is $\{0, 1\}$ or the sample space contains only two possible outcomes ($|\Omega| = 2$) (basically a coin toss)
   - If X is a Bernoulli Random Variable and let say it is used to associate the probability (p) of happening an event A, then we will map '1' for the success of the event A and map '0' for the failure of the event A. Mathematically,

$$X = \begin{cases} 1, & w.p. \;\; p \\ 0, & w.p. \;\; 1-p \end{cases}$$

   - And the PMF of X looks like,

$$p_X(x) = \begin{cases} p & \text{if } x = 1 \\ 1-p & x = 0 \\ 0 & \text{otherwise} \end{cases} \tag{12}$$

   - The expectation of Bernoulli Random Variable, $E[X] = p$
     Proof:

Since Bernoulli random variable (X) is a discrete random variable, from the definition of expectation discrete random variable we can write,

$$\mathbb{E}[X] = \sum_{x} x p_X(x) \tag{13}$$

where $p_X(x)$ is the PMF of the discrete random variable.
Using the PMF of Bernoulli (12) in (13) gives us,

$$\begin{aligned}\mathbb{E}[X] &= \sum_{x=0}^{1} x p_X(x) && (14)\\ &= 0.(1-p) + 1.p && (15)\\ &= p && (16)\end{aligned}$$

$\therefore$ The expectation of bernoulli random variable is **p**

- The variance of Bernoulli Random Variable, $Var[X] = p(1-p)$
  Proof:
  Consider the equation for variance of a random variable,

$$\begin{aligned}\mathrm{Var}[X] &= \mathbb{E}\left[(X - \mathbb{E}[X])^2\right]\\ &= \mathbb{E}\left[X^2\right] - \mathbb{E}[X]^2 && (17)\end{aligned}$$

  Using (16) ,

$$= \mathbb{E}\left[X^2\right] - p^2 \tag{18}$$

  From the property of expectation of function of a random variable,

$$\begin{aligned}&= \sum_{x=0}^{1} x^2 p_X(x) - p^2 && (19)\\ &= p - p^2 && (20)\\ &= p(1-p) && (21)\end{aligned}$$

  $\therefore$ The variance of Bernoulli Random Variable is $\boldsymbol{p(1-p)}$.
  Some Points :
  - Bernoulli distribution only has a single parameter p
  - A random experiment with exactly two possible outcomes is generally known as a **Bernoulli trial**
  - It is not necessary that X should take only 0 or 1; instead any two discrete values can do the job but it is followed as part of tradition and mathematical convenience

2. **Multinoulli Random Variable :**

- Discrete Random Variable (X) whose sample space has more than two possible outcomes ($|\Omega| > 2$).It is a generalised Bernoulli Random Variable.
- Support often taken as $\{1, ..., c\}$, if there are 'c' possible **outcomes**. Ex : A single roll of dice can give you a number in {1,2,3,4,5,6} so here c = 6.
- From ML Perspective, if we are trying to predict a class (Y) of an object where the $|Y| > 2$ (Multi-Class Classification), then Y is represented by Multinoulli Random Variable.
- Similar to Bernoulli Random Variable, we can associate the probability of occurence (p_i) of each possible outcome $i(\in \{1, .., c\})$ and the PMF of Multinoulli Random Variable would be,

$$p_X(x) = \begin{cases} p_i & \forall x = 1, 2, \ldots, c \\ 0 & \text{otherwise} \end{cases}$$

- As you can see the Mutinoulie "distribution" is parameterised with the set of probabilities $(p_1, p_2, \cdots, p_n)$.
- The expectation of Multinoulli Random Variable is,

$$E[X] = \sum_i i p_X(i)$$

Proof: It follows from the definition of expectation of discrete random variable (13)

- The variance of Multinoulli Random Variable is,

$$Var[X] = \sum_i i^2 p_X(i) - \sum_i i p_X(i)$$

Proof: It follows from the definition of variance of discrete random variable

$$\begin{aligned} \operatorname{Var}[X] &= \mathbb{E}\left[(X - \mathbb{E}[X])^2\right] \\ &= \mathbb{E}\left[X^2\right] - \mathbb{E}[X]^2 \qquad (22) \\ &= \sum_i i^2 p_X(i) - \sum_i i p_X(i) \qquad (23) \end{aligned}$$

3. **Binomial Random Variable :**

- It is a discrete random variable which counts the no. of times an event A occurs in a 'n' **independent** Bernoulli trails.
  Ex : Consider a random coin toss experiment (Bernoulli trail) which is repeated 'n' times independently, then one can observe count of no. of heads using a Binomial Random Variable (X).

- The Support of Binomial Random Variable would be {0,1,2,..,n} (Ex : No. of possible heads).
  Note : Don't confuse with Multinoulli Random Variable since the support looks similar, the support of Binomial Random Variable depend on n and represents the no .of times an event A occurs whereas in Multinoulli Random Variable support represents the category chosen.
- The PMF of Binomial Random Variable will be,

$$p_X(k) = \left(\frac{n!}{k!(n-k)!}\right) p^k (1-p)^{(n-k)}$$

  where k represents no. of times an event A occurred
  p is the probability associated to each Bernoulli Trail.
- The term $\left(\frac{n!}{k!(n-k)!}\right)$ refers to all possible combinations of getting 'k' successes out of 'n' independent Bernoulli trails.
- From this we can say a Binomial Random Variable depends on no. of independent trails **n** and probability associated with each Bernoulli **p**.
  Alternative Representation :
- We can represent Binomial Random Variable as the sum of 'n' independent and identically distributed (i.i.d) Bernoulli Random Variables.
  Mathematically it can be written as,

$$X = X_1 + X_2 + \cdots + X_n$$

  where X is a Binomial Random Variable and $\forall i, j$,
  - X_i and X_j are Independent.
  - And both X_i and X_j are from same Bernoulli **Distribution** (Identically Distributed).
- The expectation of Binomial Random Variable is

$$\mathbb{E}[X] = np$$

  Proof: Using the fact that Binomial Random Variable is a sum of i.i.d Bernoulli Random Variables, we can write,

$$\mathbb{E}[X] = \mathbb{E}[X_1 + X_2 + \cdots + X_n] \tag{24}$$

  Using the Linearity of Expectation,

$$\mathbb{E}[X] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \cdots + \mathbb{E}[X_n] \tag{25}$$
$$= np \tag{26}$$

  $\therefore$ The expectation of Binomial Random Variable is $\boldsymbol{np}$.

- The variance of Binomial Random Variable is

$$\text{Var}[X] = np(1-p)$$

Proof: Similar to the proof 3, since each Bernoulli is independent to each other we can write that,

$$\text{Var}[X] = \sum_{i=1}^{n} \text{Var}[X_i] \tag{27}$$

$$= np(1-p) \tag{28}$$

$\therefore$ The variance of Binomial Random Variable is $\boldsymbol{np(1-p)}$.

4. **Multinomial Random Vector :**

- As the name suggests it is a generalisation of Binomial Random Variable where we observe specific **set of counts** of multiple categories.
- In place of Bernoulli here we will take independent **Multinoulli** trails.
- The Support of Multinomial Random Vector will be the count of each category. So it is a Random Vector with dimensions being the no. of categories.
  Understanding the Multinomial Random vector :
- Consider 3 types of colored balls red(0), blue(1) and green(2) (Assume they are of large numbers). It is given that the probability of picking a color i is p_i. Now we are interested in finding the probability of drawing 2 red balls, 2 green balls and 1 blue balls in 5 **independent** trails.

  Solution :
  As you can see whenever you draw a ball, you are picking a random color i from the three colors (Multinoulli distribution) with $\Pr(X = i) = p_i$).
  In a particular case like below ,

  RGRBG

  The probability of drawing the balls in that sequence 02012 (2 reds, 2 greens,1 blue) will be

$$p_0 \times p_2 \times p_0 \times p_1 \times p_2 = p_0^2 p_2^2 p_1$$

  Number of all possible sequences which result in 2 red balls, 2 green balls and 1 blue ball will be

$$\frac{5!}{2!2!1!}$$

Overall probability of drawing 2 red balls, 2 green balls and 1 blue ball will then be given by

$$\left(\frac{5!}{2!2!1!}\right) p_0^2 p_2^2 p_1$$

- Using the previous discussion, we can generalise each Multinoulli to c categories and probability associated with each category i being p_i. Then for $\mathbf{n}$ independent trails, the probability of getting a sequence $(n_1, n_2, \cdots, n_c)$ will be,

$$= \left(\frac{n!}{n_1!n_2!\cdots n_c!}\right) p_1^{n_1} p_2^{n_2} \cdots p_c^{n_c}$$

  where n_i represents the count of each category i with $\sum_{i=1}^{c} n_i = n$.
- The above expression is the **Joint PMF** of Multinomial Random Vector for a given count of each categories
  $X_1 = n_1, X_2 = n_2, \cdots, X_c = n_c$ where X_i is the Random Variable which represents number of occurrences of *particular* category i.
- Generally Joint PMF represented as,

$$p_{X_1,X_2,\cdots X_c}(n_1, n_2, \cdots, n_c) = \left(\frac{n!}{n_1!n_2!\cdots n_c!}\right) p_1^{n_1} p_2^{n_2} \cdots p_c^{n_c}$$

  with Multinomial Random Vector $\mathbf{X} = (X_1, X_2, \cdots, X_c)^T$.
- It can also be seen as multivariate version of Binomial distribution as each X_i counts the occurence of ith category over n trails.
- The expectation of Multinomial Random Vector,

$$\mathbb{E}[\mathbf{X}] = \begin{pmatrix} np_1 & np_2 & \cdots & np_c \end{pmatrix}^T$$

  Proof: From the definition of expectation of random vector,

$$\mathbb{E}[\mathbf{X}] = \begin{pmatrix} \mathbb{E}[X_1] & \mathbb{E}[X_2] & \cdots \mathbb{E}[X_n] \end{pmatrix}^T \tag{29}$$

  Since in the case of Multinomial Random Vector, each component of the vector is a Binomial Random variable which counts each category occurence after $\boldsymbol{n}$ trails. We can write it as,

$$\mathbb{E}[\mathbf{X}] = \begin{pmatrix} np_1 & np_2 & \cdots & np_c \end{pmatrix}^T \tag{30}$$

  where p_i is the probability of occurence of each category i in Multinoulli trail.
- The variance of Multinomial Random Vector,

$$\mathrm{Var}[\mathbf{X}] = \begin{pmatrix} np_1(1-p_1) & np_2(1-p_2) & \cdots & np_c(1-p_c) \end{pmatrix}^T$$

  Proof: Similar to the proof above, variance of Multinomial Random Vector corresponds to the vector of individual variances of Binomial Random Variables $X_i s$.

$$\mathrm{Var}[\mathbf{X}] = \begin{pmatrix} np_1(1-p_1) & np_2(1-p_2) & \cdots & np_c(1-p_c) \end{pmatrix}^T \tag{31}$$

2.16 Cumulative Distribution Function in continuous case

As mentioned in discrete random variable, cummulative distribution function is used to describe for both continuous and discrete random variables. The definition is same as 2

$$F_X : \mathbb{R} \to [0, 1]$$
$$F_X(x) = \Pr(X \leq x)$$

And it satisfies all the four properties of CDF mentioned earlier

2.17 Probability Density Function

Probability density function (PDF) is the continuous analog of the probability mass function. It is used to describe the probability distribution of continuous random variable (X).
As compared to PMF, PDF will not evaluate the probability at each point. PDF got its name from *density* in Physics and represents the same. When probability *density* f_X is integrated over some *volume* $\mathrm{d}x$, we obtain the probability *mass*

$$\int_a^b f_X(x)\ \mathrm{d}x = \Pr(a \leq X \leq b)$$

So, basically the area under the curve $f_X(x)$ between $[a, b]$ represents the probability that X lies in $[a, b]$. It is also interesting to note that probability at a single point is **always zero** in case of continuous random variable (X). Since single point occupies **zero volume**, the probability mass will be zero.

$$\Pr(X = a) = \Pr(a \leq X \leq a) = \int_a^a f_X(x)\ \mathrm{d}x = 0$$

2.17.1 Relation b/w PDF and CDF

Consider the def 2,

$$\begin{aligned} F_X(x) &= \Pr(X \leq x) \\ &= \int_{-\infty}^{x} f_X(x)\ \mathrm{d}x \end{aligned}$$

Using leibnitz rule of integration,

$$\begin{aligned} \therefore f_X(x) &= \frac{\mathrm{d}}{\mathrm{d}x} F_x(x) \\ &= \lim_{h \to 0} \frac{F_X(x+h) - F_X(x)}{h} \\ &= \lim_{h \to 0} \frac{\Pr(x < X \leq x+h)}{h} \end{aligned}$$

This result can be interpreted as "mass per unit length" in Physics

2.17.2 Properties of PDF

- For any continuous random variable X

$$f_X(x) \geq 0 \ \forall \ x \in \mathbb{R}$$

- The area under the entire density curve is 1

$$\int_{\infty}^{\infty} f_X(x) \ \mathrm{d}x = 1$$

- However, $f_X(x)$ is not upper bounded by 1 or any number because it is not a probability
- In fact, there are distributions where the density not even finite, e.g., Dirac delta

2.18 Expectation of a Continuous Random Variable

- The expectation of a continuous random variable is defined as

$$\mathbb{E}[X] \triangleq \int_{-\infty}^{\infty} x \cdot f_X(x) \ \mathrm{d}x$$

- Similarly, the expectation of a function of a continuous random variable is defined as

$$\mathbb{E}[g(X)] \triangleq \int_{-\infty}^{\infty} g(x) \cdot f_X(x) \ \mathrm{d}x$$

2.19 Examples of Continuous Random Variable

1. **Uniform Random Variable :**
   - As the name suggests, the Uniform Random Variable X takes all the values in interval $[a, b]$ with equal probability so the PDF will be same in each point in $[a, b]$.
   - So consider $f_X(x) = k$ in $[a, b]$ then we can write,

$$\int_{-\infty}^{\infty} f_X(x)dx = 1 \tag{32}$$

$$\int_{a}^{b} f_X(x)dx = 1 \tag{33}$$

$$\int_{a}^{b} kdx = 1 \tag{34}$$

$$k(b-a) = 1 \tag{35}$$

$$k = \frac{1}{b-a} \tag{36}$$

- Therefore the PDF of Uniform Random Variable is,

$$f_X(x) = \begin{cases} \frac{1}{b-a} & x \in [a,b] \\ 0 & \text{otherwise} \end{cases} \tag{37}$$

The PDF looks like Fig :3

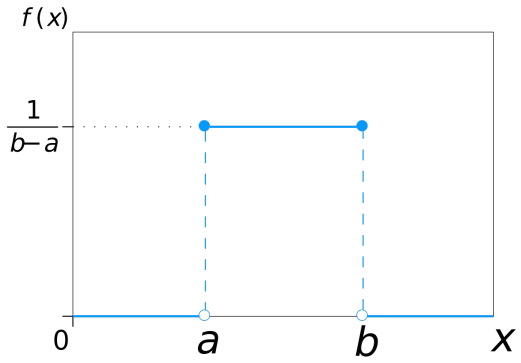


Figure 3: PDF of Uniform Random Variable

- The expectation of the uniform random variable $\mathbb{E}[X]$ is ,

$$\mathbb{E}[X] = \frac{a+b}{2}$$

Proof: Before going to the proof, it is intuitive enough that the mean is the mid point of the interval as each point is equally likely. Now using the definition of expectation of a continuous random variable,

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx \tag{38}$$

Using (37),

$$\mathbb{E}[X] = \int_a^b x \left(\frac{1}{b-a}\right) dx \tag{39}$$

$$= \frac{1}{b-a}\left[\frac{x^2}{2}\right]_a^b \tag{40}$$

$$= \frac{1}{b-a}\left(\frac{b^2-a^2}{2}\right) \tag{41}$$

$$= \frac{a+b}{2} \tag{42}$$

$\therefore$ The expectation of Uniform Random Variable defined in $[a,b]$ is $\frac{a+b}{2}$.

- The variance of the uniform random variable is,

$$\text{Var}[X] = \frac{(b-a)^2}{12}$$

Proof: Consider the variance of a random variable X,

$$\text{Var}[X] = \mathbb{E}\left[X^2\right] - (\mathbb{E}[X])^2 \tag{43}$$

Using (42),

$$\text{Var}[X] = \int_a^b 2x f_X(x)dx - \left(\frac{a+b}{2}\right)^2 \tag{44}$$

$$= \frac{1}{b-a}\int_a^b x^2 dx - \left(\frac{a+b}{2}\right)^2 \tag{45}$$

$$= \frac{1}{b-a}\left[\frac{x^3}{3}\right]_a^b - \frac{(a+b)^2}{4} \tag{46}$$

$$= \frac{b^2+a^2+ab}{3} - \frac{a^2+b^2+2ab}{4} \tag{47}$$

$$= \frac{a^2+b^2-2ab}{12} \tag{48}$$

$$= \frac{(b-a)^2}{12} \tag{49}$$

$\therefore$ The variance of the uniform random variable is $\frac{(b-a)^2}{12}$.

- The CDF of X will be

$$F_X(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & x \in [a,b] \\ 1 & x > b \end{cases}$$

Proof: Consider the CDF of the random variable X,

$$F_X(x) = \Pr(X \leq x) \tag{50}$$

We know that,

$$\Pr(a \leq X \leq b) = \int_a^b f_X(x)\,dx \tag{51}$$

Using (51) in (50),

$$F_X(x) = \int_{-\infty}^x f_X(x)\,dx \tag{52}$$

Case 1 : If $x < a$,

$$f_X(x) = 0$$
$$\implies F_X(x) = 0 \tag{53}$$

Case 2 : If $a \leq x \leq b$,

$$f_X(x) = \frac{1}{b-a} \quad x \in [a,b]$$

$$\implies F_X(x) = \int_a^x f_X(x)\,dx \tag{54}$$

$$= \int_a^x \frac{1}{b-a} dx \tag{55}$$

$$= \frac{x-a}{b-a} \tag{56}$$

Case 3 : If $x > b$,

$$f_X(x) = 0 \text{ if } x \geq b$$

$$\implies F_X(x) = \int_a^b f_X(x)\,dx + \int_b^x f_X(x)\,dx \tag{57}$$

$$F_X(x) = \int_a^b \frac{1}{b-a} dx + 0 \tag{58}$$

$$= 1 \tag{59}$$

Considering all the cases,

$$F_X(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & x \in [a,b] \\ 1 & x > b \end{cases} \tag{60}$$

2. **<u>Gaussian Random Variable :</u>**

Motivation for Gaussian Random Variable :

(a) The noise/error in the measurements are generally modelled using Gaussian random variables

(b) Thanks to the Central Limit Theorem, it plays a crucial role in **learning** from data

(c) Features which are continuous in nature are generally modelled as Gaussian Distribution

- A continuous random variable (X) is said to have Gaussian/Normal Distribution i.e., $X \sim \mathcal{N}(\mu, \sigma^2)$ if its PDF $f_X(x)$ is

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right) \quad x \in \mathbb{R}$$

- The PDF of normal random variable is a bell-shaped curve as you can see in Fig : 4 and it is symmetric about mean μ and σ signifies the spread of the bell curve.

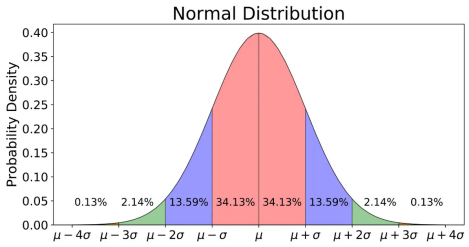


Figure 4: PDF of Gaussian Random Variable

- The Expectation of the Gaussian is μ
  Proof: Consider the expectation of the Gaussian Random Variable,

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x)\, dx \tag{61}$$

Put $t := x - \mu$

$$= \int_{-\infty}^{\infty} (t+\mu) \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-t^2}{2\sigma^2}\right) dt \tag{62}$$

$$= \int_{-\infty}^{\infty} t \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-t^2}{2\sigma^2}\right) dt \; +$$

$$\mu \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-t^2}{2\sigma^2}\right) dt \tag{63}$$

$$= 0 + \mu.1 \left(\because \; t \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-t^2}{2\sigma^2}\right) \text{is odd function}\right) \tag{64}$$

$$= \mu \tag{65}$$

$\therefore$ The Expectation of Gaussian Random Variable is μ. And it is also intuitive since the distribution is symmetric about μ (See in Figure : 4).

- The Variance of the Gaussian Random Variable is, σ^2
  Proof: Consider the variance of a continuous random variable,

$$\text{Var}[x] = \int_{-\infty}^{\infty} x^2 f_X(x)\, dx - \left(\int_{-\infty}^{\infty} x f_X(x)\, dx\right)^2 \tag{66}$$

From the above discussion,

$$\text{Var}[x] = \int_{-\infty}^{\infty} x^2 f_X(x)\, dx - \mu^2 \tag{67}$$

To evaluate the expression consider below definite integral,

$$\int_{-\infty}^{\infty} x^2 f_X(x)\,dx = \int_{-\infty}^{\infty} x^2 \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)\right\} dx \quad (68)$$

$$= \int_{-\infty}^{\infty} \left(\sqrt{2}\sigma t + \mu\right)^2 \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-t^2\right) \sqrt{2}\sigma dt \quad (69)$$

On solving,

$$= \frac{2\sigma^2}{\sqrt{\pi}} \int_{-\infty}^{\infty} t^2 \exp\left(-t^2\right) dt + \frac{\mu^2}{\sqrt{\pi}} \int_{-\infty}^{\infty} \exp\left(-t^2\right) dt + \frac{2\mu\sigma\sqrt{2}}{\sqrt{\pi}} \int_{-\infty}^{\infty} t \exp\left(-t^2\right) dt \quad (70)$$

$$= \frac{2\sigma^2}{\sqrt{\pi}} \int_{-\infty}^{\infty} t^2 \exp\left(-t^2\right) dt + \mu^2.1 + 0 \quad (71)$$

$$= \frac{2\sigma^2}{\sqrt{\pi}} \left[-t\left(\frac{\exp\left(-t^2\right)}{2}\right) + \int \exp\left(\frac{-t^2}{2}\right) dt\right]_{-\infty}^{\infty} + \mu^2 \text{(Using Integration By parts)} \quad (72)$$

$$= \frac{2\sigma^2}{\sqrt{\pi}} \int_{-\infty}^{\infty} \exp\left(\frac{-t^2}{2}\right) dt + \mu^2 \quad (73)$$

$$\implies \int_{-\infty}^{\infty} x^2 f_X(x)\,dx - \mu^2 = \frac{2\sigma^2}{\sqrt{\pi}} \frac{\sqrt{\pi}}{2} \text{ (Using PDF condition)} \quad (74)$$

$$\implies \text{Var}[X] = \sigma^2 \quad (75)$$

$\therefore$ The variance of Gaussian Random variable is $\text{Var}[X] = \sigma^2$.

- In the special case when Normal distribution has $\mu = 0$ and $\sigma = 1$ we call it as **Standard Normal**. And we can relate any Normal random variable $X \sim \mathcal{N}(\mu, \sigma^2)$ with standard normal Z using the following relationship,

$$X = \sigma Z + \mu$$

Proof: We need to prove that if $X \sim \mathcal{N}\left(\mu, \sigma^2\right)$ then Z follows Standard Normal Distribution. To do that consider the CDF of X,

$$\Pr(X \leq x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(t-\mu)^2}{2\sigma^2}\right) dt \quad (76)$$

$$\Pr(\sigma Z + \mu \leq x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(t-\mu)^2}{2\sigma^2}\right) dt \quad (77)$$

$$\Pr\left(Z \leq \frac{x-\mu}{\sigma}\right) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(t-\mu)^2}{2\sigma^2}\right) dt \quad (78)$$

Consider $z := \frac{x-\mu}{\sigma}$,

$$\Pr(Z \leq z) = \int_{-\infty}^{\mu+\sigma z} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(t-\mu)^2}{2\sigma^2}\right) dt \tag{79}$$

Now put $k = \frac{t-\mu}{\sigma}$,

$$\Pr(Z \leq z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}} \exp\left(\frac{-k^2}{2}\right) dk \tag{80}$$

Since a continuous random variable is defined by its PDF, we can clearly say that Z is a Standard Normal.

- The general result will be *linear combination of any n Normal random variables will be a Normal Random Variable* and has the following properties,

$$X = c_1X_1 + c_2X_2 + \cdots + c_nX_n$$

$E[X] = c_1\mu_1 + c_2\mu_2 + \cdots + c_n\mu_n$
$Var[X] = c_1^2\sigma_1^2 + c_2^2\sigma_2^2 + \cdots + c_n^2\sigma_n^2$

- If Z is a Standard Normal and $F_Z(z)$ being the CDF of Z then,
  - $F_Z(z) + F_Z(-z) = 1$
    Proof: Consider LHS,

$$F_Z(z) + F_Z(-z) = \int_{-\infty}^{z} f_Z(z)dz + \int_{-\infty}^{-z} f_Z(z)dz \tag{81}$$

$$= \int_{-\infty}^{z} f_Z(z)dz + \int_{t}^{\infty} f_Z(-t)dt \tag{82}$$

$$= \int_{-\infty}^{z} f_Z(z)dz + \int_{t}^{\infty} f_Z(t)dt \ (\because f_Z(z) \text{ is symmetric about } 0) \tag{83}$$

$$= 1 \tag{84}$$

  - $F_Z(0) = \frac{1}{2}$

    Proof: It is trivial, since $f_Z(z)$ is symmetric about 0.

3. **<u>Multivariate Gaussian :</u>**

- It is generalisation of a Gaussian random variable
- From the perspective of machine learning, if all the *features* in the data takes continuous values then enitre data can be considered as a **Multivariate Gaussian**

$$\mathbf{X} = \begin{pmatrix} X_1 \\ \vdots \\ X_n \end{pmatrix}$$

- We say that the random vector $\mathbf{X}$ is multivariate gaussian if the individual components are **jointly Gaussian** i.e., any linear combination of the individual components should be a Gaussian.
- A Multivariate Gaussian $\mathbf{X}$ is represented as $\mathbf{X} \sim (\boldsymbol{\mu}, \boldsymbol{\Sigma})$ where $\boldsymbol{\mu}$ is the $E[\mathbf{X}]$ and $\boldsymbol{\Sigma}$ is the **covariance matrix** of $\boldsymbol{X}$ i.e.,

$$\boldsymbol{\Sigma} = E[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^T]$$

- Then the joint density of $\mathbf{X}$ is,

$$f_{X_1, X_2, \cdots X_n}(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^n |\boldsymbol{\Sigma}|}} \exp\left(-\frac{1}{2}(\mathbf{X} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{X} - \boldsymbol{\mu})\right)$$

Note : Putting n = 1 will give the PDF of normal random variable

Special Case of Multivariate Gaussian : When each pair of (X_i, X_j) are independent then the covariance matrix $\boldsymbol{\Sigma}$ will become a **diagonal matrix** leaving out only the variances of each X_i.
Proof: Given that $X_i,\ X_j\ (i \neq j)$ are independent,

$$\begin{aligned}
\boldsymbol{\Sigma}[i,j] &= \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)] && (85)\\
&= \mathbb{E}[(X_i - \mu_i)]\,\mathbb{E}[(X_j - \mu_j)] && (86)\\
&= (\mathbb{E}[X_i] - \mu_i)(\mathbb{E}[X_j] - \mu_j) && (87)\\
&= 0 && (88)
\end{aligned}$$

Hence this results in a diagonal matrix

- In practice, when we are dealing with real data , we often assume that the above special case to be true. It is generally known as **Na ïve Bayes** assumption since it is *naïve* to assume that all the features are independent to each other.
- In the special case since each pair (X_i, X_j) independent , the Joint density looks like,

$$f_{X_1, X_2, \cdots X_n}(\mathbf{x}) = f_{X_1}(x_1) f_{X_2}(x_2) \cdots f_{X_n}(x_n)$$

2.20 Joint Distribution of a Random Vector

- The joint distribution of a random vector is the corresponding probability distribution of all possible tuples
- If $X_1, \ldots, X_n$ are discrete random variables then the joint PMF of $X_1, \ldots, X_n$ is defined as

$$p_{X_1 \ldots X_n} : \mathbb{R}^n \to [0, 1]$$
$$p_{X_1 \ldots X_n}(x_1, \ldots, x_n) = \Pr(X_1 = x_1, \ldots, X_n = x_n)$$

- If $X_1, \ldots, X_n$ are random variables then the joint CDF of $X_1, \ldots, X_n$ is defined as

$$F_{X_1 \ldots X_n} : \mathbb{R}^n \to [0, 1]$$
$$F_{X_1 \ldots X_n}(x_1, \ldots, x_n) = \Pr\left(X_1 \leq x_1, \ldots, X_n \leq x_n\right)$$

- If additionally, the random vector is continuous, i.e., $X_1, \ldots, X_n$ are continuous random variables, then the joint PDF is defined as

$$f_{X_1 \ldots X_n} : \mathbb{R}^n \to \mathbb{R}_{\geq 0}$$
$$f_{X_1 \ldots X_n}(x_1, \ldots, x_n) = \frac{\partial^n}{\partial x_1 \cdots \partial x_n} F_{X_1 \ldots X_n}(x_1, \ldots, x_n)$$

2.20.1 Properties of the joint

If p_{XY} is a joint PMF, them

1. $$0 \leq p_{XY}(x, y) \leq 1 \quad \forall\ x, y \in \mathbb{R}$$

2. $$\sum_x \sum_y p_{XY}(x, y) = 1$$

3. $$\mathbb{E}\left[g(X, Y)\right] \triangleq \sum_x \sum_y g(x, y) \cdot p_{XY}(x, y)$$

If f_{XY} is a joint PDF, them

1. $$\int_a^b \int_c^d f_{XY}(x, y)\, \mathrm{d}x\, \mathrm{d}y = \Pr\left(a \leq X \leq b, c \leq Y \leq d\right)$$

2. $$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y)\, \mathrm{d}x\, \mathrm{d}y = 1$$

3. $$\mathbb{E}\left[g(X, Y)\right] \triangleq \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) \cdot p_{XY}(x, y)\, \mathrm{d}x\, \mathrm{d}y$$

Note that for defining the joint, the random variables in the random vector must either be all continuous or all discrete. **Joint distribution of a discrete and a continuous random variable is not defined**

2.20.2 Example

Let $X \in \mathcal{X} = \{0, 1\}$ denote the outcome of a fair coin toss and $Y \in \mathcal{Y} = \{1, 2, 3, 4, 5, 6\}$ denote the outcome of a fair dice roll. Assume that the coin toss and dice roll are independent of each other. Find the joint distribution.

Solution:

The joint distribution p_{XY} will be defined on $\mathcal{X} \times \mathcal{Y}$ which has a total of $2 \times 6 = 12$ elements

For each of these elements (x, y) where $x \in \mathcal{X}$ and $y \in \mathcal{Y}$, we have

$$\begin{aligned} p_{XY}(x, y) &= \Pr(X = x, Y = y) && (89)\\ &= \Pr(X = x) \Pr(Y = y) \qquad \text{by independence} && (90)\\ &= \frac{1}{2} \cdot \frac{1}{6} && (91)\\ &= \frac{1}{12} && (92) \end{aligned}$$

2.21 Marginal Distribution

- The individual probability distributions involved in a join are called as the marginals of the joint
- Say F_{XY} is a joint distribution, then F_X and F_Y are said to be the marginal distributions of F_{XY}
- Given a joint distribution, the marginals can be uniquely determined using a process known as **marginalization**

$$F_X(x) = \lim_{y \to \infty} F_{XY}(x, y) \tag{93}$$

$$p_X(x) = \sum_y p_{XY}(x, y) \tag{94}$$

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y)\, \mathrm{d}y \tag{95}$$

2.21.1 Example

In the above example, we have computed the joint PMF

We can now obtain the marginal PMFs of X and Y

$$p_X(x) = \sum_y p_{XY}(x, y) \tag{96}$$

$$= \sum_{y=1}^{6} \frac{1}{12} \tag{97}$$

$$= \frac{1}{2} \tag{98}$$

$$p_Y(y) = \sum_x p_{XY}(x, y) \tag{99}$$

$$= \sum_{x=0}^{1} \frac{1}{12} \tag{100}$$

$$= \frac{1}{6} \tag{101}$$

which are exactly what we started with

2.22 Conditional Distribution

- If X and Y are both discrete random variables, then the conditional PMF of X given Y is defined as

$$p_{Y|X}(y \,|\, x) = \frac{p_{XY}(xy)}{p_X(x)} \tag{102}$$

- If X and Y are both continuous random variables, then the conditional PDF of X given Y is defined as

$$f_{Y|X}(y \,|\, x) = \frac{f_{XY}(xy)}{f_X(x)} \tag{103}$$

- If X is continuous and Y is discrete, then the conditional CDF and PMF of Y given X are defined as

$$F_{Y|X}(y \,|\, x) = \lim_{h \to 0} \Pr\left(Y \leq y \,|\, x \leq X \leq x + h\right) \tag{104}$$

$$p_{Y|X}(y \,|\, x) = \lim_{h \to 0} \Pr\left(Y = y \,|\, x \leq X \leq x + h\right) \tag{105}$$

$$= \lim_{h \to 0} \frac{\Pr\left(Y = y,\, x \leq X \leq x + h\right)}{\Pr\left(x \leq X \leq x + h\right)} \tag{106}$$

We consider a small interval $(x, x + h)$ because at a specific point, the probability of a continuous random variable is zero which would have caused a problem in the denominator

- If X is continuous and Y is discrete, then the conditional CDF and PDF of X given Y are defined as

$$F_{X|Y}(x \,|\, y) = \Pr(X \leq x \,|\, Y = y) \tag{107}$$

$$= \frac{\Pr(X \leq x, Y = y)}{\Pr(Y = y)} \tag{108}$$

$$f_{X|Y}(x \,|\, y) = \frac{\mathrm{d}}{\mathrm{d}x} F_{X|Y}(x \,|\, y) \tag{109}$$

2.23 Independence of Random Variables

- Two random variables X and Y are said to be independent if and only if

$$F_{XY}(x, y) = F_X(x)\, F_Y(y)$$

or equivalently

$$p_{XY}(x, y) = p_X(x)\, p_Y(y) \qquad \text{for discrete} \tag{110}$$

$$f_{XY}(x, y) = f_X(x)\, f_Y(y) \qquad \text{for continuous} \tag{111}$$

- If X and Y are independent random variables, g is a function of only X (and not Y) and h is a function of only Y (and not X), then

$$\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\,\mathbb{E}[h(Y)] \tag{112}$$

- However, note that $\mathrm{Var}[XY] = \mathrm{Var}[X]\,\mathrm{Var}[Y]$ is not true in general even if they are independent

- X and Y are said to be *conditionally independent* given Z if

$$p_{XY|Z}(x, y \,|\, z) = p_{X|Z}(x \,|\, z)\, p_{Y|Z}(y \,|\, z) \qquad \text{for discrete} \tag{113}$$

$$f_{XY|Z}(x, y \,|\, z) = f_{X|Z}(x \,|\, z)\, f_{Y|Z}(y \,|\, z) \qquad \text{for continuous} \tag{114}$$

- Random variables $X_1, \ldots, X_n$ are said to be **independent and identically distributed** (iid) if they have the same distribution and are mutually independent

2.24 Bayes' Rule for Events

If $A, B \subseteq \Omega$ are events, then Bayes' rule states that

$$\Pr(A \,|\, B) = \frac{\Pr(B \,|\, A)\ \Pr(A)}{\Pr(B)}$$

Proof:

By the definition of conditional probability

$$\Pr(A \mid B) = \frac{\Pr(A, B)}{\Pr(B)} \tag{115}$$

$$= \frac{\Pr(A, B)}{\Pr(A)} \cdot \frac{\Pr(A)}{\Pr(B)} \tag{116}$$

$$= \Pr(B \mid A) \cdot \frac{\Pr(A)}{\Pr(B)} \tag{117}$$

2.25 Bayes' Rule for Random Variables

If X and Y are random variables of any combination of types then we have

$$p(X \mid Y) = \frac{p(Y \mid X)p(X)}{p(Y)}$$

where p denotes the respective distribution based on the type of the random variable

Proof:

CASE 1: Both X and Y are discrete

p here simply represents the probability mass function and can directly be written in terms of probabilities

$$p_{Y|X}(y \mid x) = \frac{p_{XY}(x, y)}{p_X(x)} \tag{118}$$

$$= \frac{\Pr(X = x, Y = y)}{\Pr(X = x)} \tag{119}$$

$$= \frac{\Pr(X = x, Y = y)}{\Pr(Y = y)} \cdot \frac{\Pr(Y = y)}{\Pr(X = x)} \tag{120}$$

$$= p_{X|Y}(x \mid y) \cdot \frac{p_Y(y)}{p_X(x)} \tag{121}$$

$$\therefore p(y \mid x) = \frac{p(x \mid y)p(y)}{p(x)} \tag{122}$$

assuming $\Pr(Y = y)$ and $\Pr(X = x)$ are non-zero

CASE 2: X and Y are jointly continuous

p in this case represents the probability density function

Consider the cumulative distribution function of $Y \mid X$

$$F_{Y|X}(y \mid x) = \lim_{h \to 0} \Pr(Y \leq y \mid x \leq X \leq x + h) \tag{123}$$

We are considering an infinitesimal interval around x because for a continuous random variable X

$$\Pr(X = x) = 0 \tag{124}$$

Now

$$F_{Y|X}(y \mid x) = \lim_{h \to 0} \frac{\Pr\left(Y \leq y,\, x \leq X \leq x + h\right)}{\Pr\left(x \leq X \leq x + h\right)} \tag{125}$$

$$= \lim_{h \to 0} \frac{\displaystyle\int_{-\infty}^{y} \int_{x}^{x+h} f_{XY}(u, v) \mathrm{d}u \mathrm{d}v}{\displaystyle\int_{x}^{x+h} f_X(u) \mathrm{d}u} \tag{126}$$

where $f_{XY}(x, y)$ is the joint probability density function of X and Y whose marginal density functions are $f_X(x)$ and $f_Y(y)$ respectively
By the mean value theorem for integrals, $\exists c \in (x, x + h)$ such that

$$\int_{x}^{x+h} f_X(u) \mathrm{d}u = ((x + h) - (x)) f_X(c) = h f_X(c) \tag{127}$$

But as $h \to 0$

$$\lim_{h \to 0} f_X(c) = f_X(x) \tag{128}$$

since $c \in (x, x + h)$

$$\therefore \int_{x}^{x+h} f_X(u) \mathrm{d}u \approx h f_X(x) \tag{129}$$

Similarly

$$\int_{x}^{x+h} f_{XY}(u, v) \mathrm{d}u \approx h f_{XY}(x, v) \tag{130}$$

Thus

$$F_{Y|X}(y \mid x) = \lim_{h \to 0} \frac{\displaystyle\int_{-\infty}^{y} h f_{XY}(x, v) \mathrm{d}v}{h f_X(x)} \tag{131}$$

$$= \lim_{h \to 0} \frac{h \displaystyle\int_{-\infty}^{y} f_{XY}(x, v) \mathrm{d}v}{h f_X(x)} \tag{132}$$

$$= \int_{-\infty}^{y} \frac{f_{XY}(x, v)}{f_X(x)} \mathrm{d}v \tag{133}$$

since h and $f_X(x)$ are constant with respect to v
Now, the probability density function of $Y \mid X$ is given by

$$f_{Y|X}(y \mid x) = \frac{\mathrm{d}}{\mathrm{d}y} F_{Y|X}(y \mid x) \tag{134}$$

$$= \frac{\mathrm{d}}{\mathrm{d}y} \int_{-\infty}^{y} \frac{f_{XY}(x, v)}{f_X(x)} \mathrm{d}v \tag{135}$$

$$= \frac{f_{XY}(x, y)}{f_X(x)} \tag{136}$$

Similarly

$$f_{X|Y}(x \mid y) = \frac{f_{XY}(x, y)}{f_Y(y)} \tag{137}$$

assuming $f_X(x)$ and $f_Y(y)$ are non-zero
From (137)

$$f_{XY}(x, y) = f_{X|Y}(x \mid y) f_Y(y) \tag{138}$$

Substituting in (136), we get

$$f_{Y|X}(y \mid x) = \frac{f_{X|Y}(x \mid y) f_Y(y)}{f_X(x)} \tag{139}$$

$$\therefore p(y \mid x) = \frac{p(x \mid y) p(y)}{p(x)} \tag{140}$$

CASE 3: One of X and Y is continuous and the other is discrete

Without loss of generality, let us assume that X is continuous and Y is discrete

There is no loss of generality because Bayes' rule is symmetric in X and Y

Note that X and $X \mid Y$ have probability density functions, and Y and $Y \mid X$ have probability mass functions
Consider the probability mass function of $Y \mid X$

$$p_{Y|X}(y \mid x) = \lim_{h \to 0} \Pr(Y = y \mid x \leq X \leq x + h) \tag{141}$$

$$= \lim_{h \to 0} \frac{\Pr(Y = y,\, x \leq X \leq x + h)}{\Pr(x \leq X \leq x + h)} \tag{142}$$

Now

$$\Pr(x \leq X \leq x + h \mid Y = y) = \frac{\Pr(Y = y,\, x \leq X \leq x + h)}{\Pr(Y = y)} \tag{143}$$

$$\implies \Pr(Y = y,\, x \leq X \leq x + h) = \Pr(x \leq X \leq x + h \mid Y = y) \Pr(Y = y) \tag{144}$$

This can be written whenever $\Pr(Y = y) \neq 0$

$\Pr(Y = y)$ is not always zero because Y is discrete
Thus

$$p_{Y|X}(y \mid x) = \lim_{h \to 0} \frac{\Pr(x \leq X \leq x + h \mid Y = y) \Pr(Y = y)}{\Pr(x \leq X \leq x + h)} \tag{145}$$

$$= \lim_{h \to 0} \frac{\left(\int_x^{x+h} f_{X|Y}(u \mid y) \mathrm{d}u \right) p_Y(y)}{\int_x^{x+h} f_X(u) \mathrm{d}u} \tag{146}$$

Using the mean value theorem again, we get

$$\int_x^{x+h} f_X(u)\mathrm{d}u \approx h f_X(x) \tag{147}$$

$$\int_x^{x+h} f_{X|Y}(u \,|\, y)\mathrm{d}u \approx h f_{X|Y}(x \,|\, y) \tag{148}$$

Substituting back, we get

$$p_{Y|X}(y \,|\, x) = \lim_{h\to 0} \frac{h f_{X|Y}(x \,|\, y) p_Y(y)}{h f_X(x)} \tag{149}$$

$$= \frac{f_{X|Y}(x \,|\, y) p_Y(y)}{p_X(x)} \tag{150}$$

$$\therefore p(y \,|\, x) = \frac{p(x \,|\, y)p(y)}{p(x)} \tag{151}$$

assuming $p_X(x)$ is non-zero

We have already assumed $p_Y(y)$ is non-zero when writing (143)

2.26 Total Expectation Rule

If X and Y are random variables of any combination of types then we have

$$E[f(X,Y)] = E\left[E\left[f(X,Y) \,|\, X\right]\right] = E\left[E\left[f(X,Y) \,|\, Y\right]\right]$$

where f(X,Y) is any function of X and Y

Proof:

CASE 1: Both X and Y are discrete

For a fixed $X = x$, consider the expectation of $f(X,Y) \,|\, X$

$$\mathbb{E}\left[f(X,Y) \,|\, X = x\right] = \sum_y f(x,y) p_{Y|X}(y \,|\, x) \tag{152}$$

where p is a probability mass function

This is a function of x, so denote it by

$$g(x) = \mathbb{E}\left[f(X,Y) \,|\, X = x\right] \tag{153}$$

Now, $\mathbb{E}\left[f(X,Y) \,|\, X\right]$ is a random variable of X. Denote this by

$$g(X) = \mathbb{E}\left[f(X,Y) \,|\, X\right] \tag{154}$$

Thus

$$\mathbb{E}\left[\mathbb{E}\left[f(X,Y) \,|\, X\right]\right] = \mathbb{E}\left[g(X)\right] \tag{155}$$

$$= \sum_x g(x) p_X(x) \tag{156}$$

$$= \sum_x \mathbb{E}\left[f(X,Y) \,|\, X = x\right] p_X(x) \tag{157}$$

On substituting (152) into this equation, we get

$$\mathbb{E}\left[\mathbb{E}\left[f(X,Y)\,|\,X\right]\right] = \sum_x \left(\sum_y f(x,y) p_{Y|X}(y\,|\,x)\right) p_X(x) \tag{158}$$

$$= \sum_x \sum_y f(x,y) p_{Y|X}(y\,|\,x) p_X(x) \tag{159}$$

$$= \sum_x \sum_y f(x,y) \frac{p_{XY}(x,y)}{p_X(x)} p_X(x) \tag{160}$$

$$= \sum_x \sum_y f(x,y) p_{XY}(x,y) \tag{161}$$

$$= \mathbb{E}\left[f(X,Y)\right] \tag{162}$$

since for discrete random variables

$$p_{Y|X}(y\,|\,x) = \frac{p_{XY}(x,y)}{p_X(x)} \tag{163}$$

Similarly, we can show that

$$\mathbb{E}\left[\mathbb{E}\left[f(X,Y)\,|\,Y\right]\right] = \mathbb{E}\left[f(X,Y)\right] \tag{164}$$

Therefore, from (162) and (164)

$$\mathbb{E}\left[f(X,Y)\right] = \mathbb{E}\left[\mathbb{E}\left[f(X,Y)\,|\,X\right]\right] = \mathbb{E}\left[\mathbb{E}\left[f(X,Y)\,|\,Y\right]\right] \tag{165}$$

CASE 2: X and Y are jointly continuous

For a fixed $X = x$, consider the expectation of $f(X,Y)\,|\,X$

$$\mathbb{E}\left[f(X,Y)\,|\,X = x\right] = \int_{-\infty}^{\infty} f(x,y) p_{Y|X}(y\,|\,x) \mathrm{d}y \tag{166}$$

where p is a probability density function
This is a function of x, so denote it by

$$g(x) = \mathbb{E}\left[f(X,Y)\,|\,X = x\right] \tag{167}$$

Now, $\mathbb{E}\left[f(X,Y)\,|\,X\right]$ is a random variable of X. Denote this by

$$g(X) = \mathbb{E}\left[f(X,Y)\,|\,X\right] \tag{168}$$

Thus

$$\mathbb{E}\left[\mathbb{E}\left[f(X,Y)\,|\,X\right]\right] = \mathbb{E}\left[g(X)\right] \tag{169}$$

$$= \int_{-\infty}^{\infty} g(x) p_X(x) \mathrm{d}x \tag{170}$$

$$= \int_{-\infty}^{\infty} \mathbb{E}\left[f(X,Y)\,|\,X = x\right] p_X(x) \mathrm{d}x \tag{171}$$

On substituting (166) into this equation, we get

$$\mathbb{E}\left[\mathbb{E}\left[f(X,Y)\,|\,X\right]\right] = \int_{-\infty}^{\infty}\left(\int_{-\infty}^{\infty} f(x,y)p_{Y|X}(y\,|\,x)\mathrm{d}y\right)p_X(x)\mathrm{d}x \tag{172}$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)p_{Y|X}(y\,|\,x)p_X(x)\mathrm{d}x\mathrm{d}y \tag{173}$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\frac{p_{XY}(x,y)}{p_X(x)}p_X(x)\mathrm{d}x\mathrm{d}y \tag{174}$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)p_{XY}(x,y)\mathrm{d}x\mathrm{d}y \tag{175}$$

$$= \mathbb{E}\left[f(X,Y)\right] \tag{176}$$

since for continuous random variables we have proved in (136) that

$$p_{Y|X}(y\,|\,x) = \frac{p_{XY}(x,y)}{p_X(x)} \tag{177}$$

Similarly, we can show that

$$\mathbb{E}\left[\mathbb{E}\left[f(X,Y)\,|\,Y\right]\right] = \mathbb{E}\left[f(X,Y)\right] \tag{178}$$

Therefore, from (176) and (178)

$$\mathbb{E}\left[f(X,Y)\right] = \mathbb{E}\left[\mathbb{E}\left[f(X,Y)\,|\,X\right]\right] = \mathbb{E}\left[\mathbb{E}\left[f(X,Y)\,|\,Y\right]\right] \tag{179}$$

CASE 3: One of X and Y is continuous and the other is discrete

Without loss of generality, let us assume that X is continuous and Y is discrete

Note that X and $X\,|\,Y$ have probability density functions, and Y and $Y\,|\,X$ have probability mass functions
For a fixed $X = x$, consider the expectation of $f(X,Y)\,|\,X$

$$\mathbb{E}\left[f(X,Y)\,|\,X = x\right] = \sum_{y} f(x,y)p_{Y|X}(y\,|\,x) \tag{180}$$

This is a function of x, so denote it by

$$g(x) = \mathbb{E}\left[f(X,Y)\,|\,X = x\right] \tag{181}$$

Now, $\mathbb{E}\left[f(X,Y)\,|\,X\right]$ is a random variable of X. Denote this by

$$g(X) = \mathbb{E}\left[f(X,Y)\,|\,X\right] \tag{182}$$

Thus

$$\mathbb{E}\left[\mathbb{E}\left[f(X,Y)\,|\,X\right]\right] = \mathbb{E}\left[g(X)\right] \tag{183}$$

$$= \int_{-\infty}^{\infty} g(x)p_X(x)\mathrm{d}x \tag{184}$$

$$= \int_{-\infty}^{\infty} \mathbb{E}\left[f(X,Y)\,|\,X=x\right]p_X(x)\mathrm{d}x \tag{185}$$

On substituting (180) into this equation, we get

$$\mathbb{E}\left[\mathbb{E}\left[f(X,Y)\,|\,X\right]\right] = \int_{-\infty}^{\infty}\left(\sum_y f(x,y)p_{Y|X}(y\,|\,x)\right)p_X(x)\mathrm{d}x \tag{186}$$

$$= \sum_y\left(\int_{-\infty}^{\infty} f(x,y)p_{Y|X}(y\,|\,x)p_X(x)\mathrm{d}x\right) \tag{187}$$

$$= \mathbb{E}\left[f(X,Y)\right] \tag{188}$$

Now, fix a $Y=y$
Consider the expectation of $f(X,Y)\,|\,Y$

$$\mathbb{E}\left[f(X,Y)\,|\,Y=y\right] = \int_{-\infty}^{\infty} f(x,y)p_{X|Y}(x\,|\,y)\mathrm{d}x \tag{189}$$

This is a function of y, so denote it by

$$h(y) = \mathbb{E}\left[f(X,Y)\,|\,Y=y\right] \tag{190}$$

Now, $\mathbb{E}\left[f(X,Y)\,|\,Y\right]$ is a random variable of Y. Denote this by

$$h(Y) = \mathbb{E}\left[f(X,Y)\,|\,Y\right] \tag{191}$$

Thus

$$\mathbb{E}\left[\mathbb{E}\left[f(X,Y)\,|\,Y\right]\right] = \mathbb{E}\left[h(Y)\right] \tag{192}$$

$$= \sum_y h(y)p_y(y) \tag{193}$$

$$= \sum_y \mathbb{E}\left[f(X,Y)\,|\,Y=y\right]p_Y(y) \tag{194}$$

On substituting (189) into this equation, we get

$$\mathbb{E}\left[\mathbb{E}\left[f(X,Y)\,|\,X\right]\right] = \sum_y\left(\int_{-\infty}^{\infty} f(x,y)p_{X|Y}(x\,|\,y)\mathrm{d}x\right)p_Y(y) \tag{195}$$

$$= \sum_y\left(\int_{-\infty}^{\infty} f(x,y)p_{X|Y}(x\,|\,y)p_Y(y)\mathrm{d}x\right) \tag{196}$$

By Bayes' rule

$$p_{Y|X}(y \mid x) = \frac{p_{X|Y}(x \mid y)p_Y(y)}{p_X(x)} \tag{197}$$

$$\implies p_{X|Y}(x \mid y)p_Y(y) = p_{Y|X}(y \mid x)p_X(x) \tag{198}$$

Thus

$$\mathbb{E}\left[\mathbb{E}\left[f(X,Y) \mid X\right]\right] = \sum_y \left(\int_{-\infty}^{\infty} f(x,y)p_{Y|X}(y \mid x)p_X(x)\mathrm{d}x\right) \tag{199}$$

$$= \mathbb{E}\left[f(X,Y)\right] \tag{200}$$

Therefore, from (188) and (200)

$$\mathbb{E}\left[f(X,Y)\right] = \mathbb{E}\left[\mathbb{E}\left[f(X,Y) \mid X\right]\right] = \mathbb{E}\left[\mathbb{E}\left[f(X,Y) \mid Y\right]\right] \tag{201}$$

2.26.1 Example

Suppose that the expected number of accidents per week on the highway through IIT-H is **4**. Suppose also that the numbers of commuters injured in each accident are independent random variables with a common mean of **2**. Assume also that the number of commuters injured in each accident is independent of the number of accidents that occur. What is the expected number of injuries during a week on that highway? Compute this formally using the *total expectation rule*.

Solution:

Let the number of accidents happening per week be denoted by the random variable X

It is given that

$$\mathbb{E}\left[X\right] = 4 \tag{202}$$

Let the number of commuters injured in the ith accident be denoted by the random variable Y_i where $i = 1, 2, \ldots, X$

It is also given that

$$\mathbb{E}\left[Y_i\right] = 2 \qquad \forall i = 1, 2, \ldots, X \tag{203}$$

The total number of injuries during a week is then given by another random variable Y

$$Y = Y_1 + Y_2 + \cdots Y_X = \sum_{i=1}^{X} Y_i \tag{204}$$

We have to compute the expected number of injuries during a week, i.e.,

$$\mathbb{E}\left[\sum_{i=1}^{X} Y_i\right] = \mathbb{E}\left[Y\right] \tag{205}$$

By the total expectation rule

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]] \tag{206}$$

For computing this, first fix an $X = x$

$$\begin{aligned}
\mathbb{E}[Y \mid X = x] &= \mathbb{E}\left[\sum_{i=1}^{x} Y_i\right] && \text{by the independence of } X \text{ and } Y_i && (207)\\
&= \sum_{i=1}^{x} \mathbb{E}[Y_i] && \text{by linearity of expectation} && (208)\\
&= \sum_{i=1}^{x} 2 && \text{from (203)} && (209)\\
&= 2x && && (210)
\end{aligned}$$

This is a function of x, so denote it by

$$g(x) = \mathbb{E}[Y \mid X = x] = 2x \tag{211}$$

Now, $\mathbb{E}[Y \mid X]$ is a random variable of X. Denote this by

$$g(X) = \mathbb{E}[Y \mid X] = 2X \tag{212}$$

Thus

$$\begin{aligned}
\mathbb{E}[\mathbb{E}[Y \mid X]] &= \mathbb{E}[g(X)] && (213)\\
&= \mathbb{E}[2X] && (214)\\
&= 2\mathbb{E}[X] && (215)\\
&= 2 \times 4 && (216)\\
&= 8 && (217)
\end{aligned}$$

Therefore, the expected number of injuries during a week on that highway is

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]] = 8 \tag{218}$$

2.27 Exercises on Probability

2.27.1 Probability and Set Theory

1. If A and B are sets
   (a) Give a counterexample to show that $(A \setminus B) \cup B = A$ is not always true
   (b) Prove that $(A \setminus B) \cup B = A \cup B$
   (c) Simplify $(A \,\Delta\, B) \cup (A \cap B)$

2. Give an example to demonstrate why Kolmogorov's Axiom 3 is not true for an uncountable sequence of disjoint events. (Hint: Consider $\Omega = [0, 1]$)

3. Using the Kolmogorov axioms, prove that

$$\begin{aligned}\Pr(A \cup B \cup C) = \Pr(A) + \Pr(B) + \Pr(C) \\ - \Pr(A \cap B) - \Pr(B \cap C) - \Pr(A \cap C) + \Pr(A \cap B \cap C)\end{aligned}$$

4. Consider events $A, B, C \subseteq \Omega$. Express the following events in set notation

   (a) None of the events occur

   (b) All of the events occur

   (c) At least one of the events occur

   (d) At least two of the events occur

   (e) Exactly one of the events occurs

   (f) Exactly two of the events occur

5. Prove that the Cartesian product of two countable sets is countable. Generalize this to finitely many countable sets. (Hint: Use a similar argument to the one we used to prove the countability of the rationals)

6. Is the power set of the naturals, i.e., $\mathcal{P}(\mathbb{N})$ countable?

2.27.2 Discrete Random Variables

1. Consider two random variables X and Y with X being a Bernoulli random variable with parameter p and Y is given by

$$Y = aX + b$$

   where a and b are constants

   (a) What is the support of Y?

   (b) Find the mean and variance of the random variable Y

   (c) Calculate the PMF and CDF of the random variable Y

   (d) What type of random variable is Y?

2. Consider a fair dice. We are interested in conducting a random experiment where the dice is rolled until a number greater than 2 appears. Let's denote the number of times the dice needed to be rolled by a random variable X. Find the PMF of X for $x = 1, 2, 3, \ldots$
   X is known as a **geometric random variable**.

3. A discrete random variable X is said to follow **Poisson distribution** if the PMF of the random variable X is of the form

$$\Pr(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

where $\lambda > 0$ is a parameter and $k \in \mathbb{W}$

   (a) Find $\Pr(X \leq 3)$

   (b) Find the mean and variance of a Poisson random variable X

   (c) Read up on the Poisson distribution and give a daily life example where you can model a situation with a Poisson Random variable

2.27.3 Continuous Random Variables

1. Consider a continuous random variable X which has the following PDF

$$f_X(x) = \begin{cases} ce^{-3x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

   (a) What is the value of c? (Hint: Use the properties of a PDF)

   (b) Find the probability that if I take a random sample from X, it lies in $[-2, 3]$

   (c) Find the CDF of X

   (d) Find the mean of X

2. We know that if X is a continuous random variable, $\Pr(X = c) = 0$ for any $c \in \mathcal{R}$. Apart from proving it mathematically, justify it by considering it as a random experiment.

3. Consider a random variable X with the following CDF

$$F_X(x) = \begin{cases} 0 & x < 0 \\ \frac{x^2}{2} & 0 \leq x < \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \leq x < 3 \\ 1 & x \geq 3 \end{cases}$$

   Find the PDF of X. Check whether it satisfies the properties of a PDF.

2.27.4 Joint and Marginal Distribution

1. Consider a random vector $\mathbf{Z} = (X, Y)^T$ where X and Y are standard normal random variables. Then evaluate the joint density of $\mathbf{Z}$ when,

   (a) Both are independent

(b) Both are dependent with correlation coefficient being ρ[1] $\in (-1, 1)$

2. Consider continuous random variables X and Y whose joint PDF is given by

$$f_{XY}(x, y) = \begin{cases} 5e^{-(x+5y)} & x, y \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

(a) Verify if this is a valid joint PDF

(b) Find the marginals f_X and f_Y

(c) Are X and Y independent?

(d) Find $\mathbb{E}[Y \mid X > 5]$

(e) Find $\Pr(X < Y)$

2.27.5 Bayes's Rule and Total Expectation Rule

1. Let X and Y be two independent $N(0, 1)$ random variables, and $Z = X + Y$.

(a) What is the PDF of Z

(b) Find the conditional PDF of Z given $X = x$, $f_{Z|X}(z|x)$

(c) Find the conditional PDF of X given $Z = z$, $f_{X|Z}(x|z)$

(d) Find the $\mathbb{E}[X|Z = z]$ and $\text{Var}[Z|X = x]$

2. A coin, having probability p of coming up heads, is to be successively flipped until the first head appears(Geometric Distribution). What is the expected number of flips required ?.

[1] The correlation coeffcient of two random variables X and Y is an extension of covariance and it equals to

$$\rho = \frac{\textbf{Cov}(X, Y)}{\sqrt{\text{Var}[X]\,\text{Var}[Y]}}$$

3 Statistics

3.1 Examples of datatypes

1. Nominal data:- A class of items, for example **three fruits, Apple , Orange and Strawberry**. They are not of numeric type and also you cannot rank them on any basis. Hence it comes under Nominal data. Another example would be your **blood groups** (A/B/AB/O)

2. Ordinal data:- Unlike Nominal data, you can rank the classes in a specific order. For instance, **movie ratings** (being 1 to 5), these numbers are only for representative purposes, and they cannot be modified in any manner. Another simple example would be the **grades**, from A to F.

3. Continuous variables:- They can take any value between a particular range. Some simple examples would be **height**, **weight**, **mass**,**volume**.

4. Discrete Variables:- The variables can only take specific values, some examples being **days in a week**, **school attendances**, these numbers will not be fractional, but however they can be used for calculations unlike Ordinal data.

3.2 Uses of The central tendency measures

1. When the data has numerical type values, and the distribution of the data has no outliers, then it would be best to use the mean as the centre of distribution.

2. In the case where there is an outlier (Extremely large or small value), the value might have a significant effect on the mean of the data.

3. For Categorical Data, mode can be used to find the class that is the most common in the dataset.

3.3 Boxplot

For a given 1-D data, after sorting in ascending order. The median would be represented by a line within the box, and the edges of the box being the 1st(Q1) and 3rd(Q3) quartile and the box representing the interquartile range (IQR).
the 1st and 3rd quartile represent the 25 percentile and 75 percentile, the interquartile range contains the values between them.
The minimum and the maximum equals to the $Q1 - 1.5 * IQR$ and $Q3 + 1.5 * IQR$,

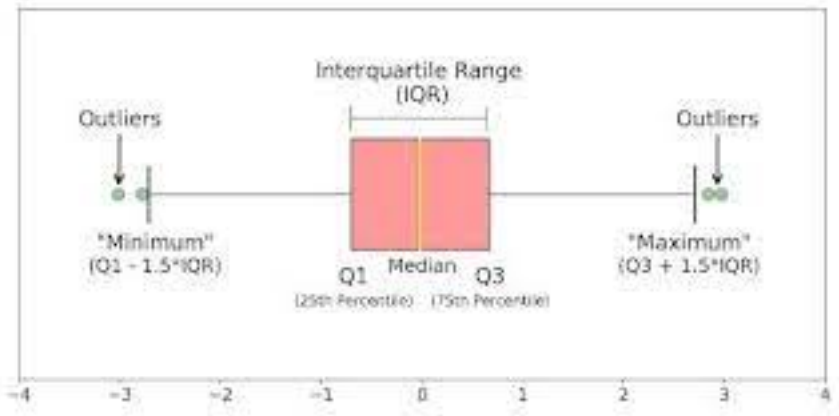


Figure 5: Boxplot

3.4 Priori and Posteriori

For any experiment, the priori is defined as the data available before analyzing. The posteriori is defined as the conclusions obtained after analyzing the data. In an experiment, where there are five different metal samples each with a different moduli, each of them are heat-treated and as a result their moduli readings give a different value.
We would like to know if the heat treatment would have a significant change in the moduli, based on the results from existing data.
What would be the priori and posteriori in this case?

3.5 Estimation of Means

Given a large data (which we call it as the population), consisting of N values, having mean as μ and standard deviation σ.
We take a subset of this data having n values, the mean of this subset being X and standard deviation s. We can say that as n approaches N ,X will be a better estimate of μ (until n=N) for the subset.
On the other hand, if we repeatedly sample (replicating) the population (without bias), the samples also having the same size 'n' and means x_i. If we plot the distribution of x_i, the figure will resemble a gaussian distribution (the more replicates, the more resemblance the distribution will be).
In addition to this, some of the characteristics of the distribution will be:-

- The mean of the distribution will be the mean of the population (μ)
- The standard deviation of the distribution will be equal to

$$S.D = \sigma/\sqrt{n}$$

An example of this would be, suppose you want to take up a survey of the glucose level of people in hyderabad, varying with their ages and try to estimate a mean for each and every age category. Since the population in this case is very large, one can take samples of people from a few localities and then draw the distribution of the means to get an estimate of the population mean. By taking more and more samples a better estimate can be obtained. Remember, it is only an estimate of the actual value, not the actual value itself.
The goal of any supervised machine learning algorithm is to best estimate the mapping function (f) for the output variable (Y) given the input data (X). The mapping function is often called the target function because it is the function a given supervised machine learning algorithm aims to approximate.

$$y = f(x) + \epsilon$$

where $\mathbb{E}(\epsilon)$=0, $\text{var}(\epsilon) = \sigma_\epsilon^2$
The prediction error for any machine learning algorithm can be broken down into three parts:

1. Bias
2. Variance
3. Irreducible error

$$\begin{aligned}\mathbb{E}[(y - \hat{f}(x))^2] &= \mathbb{E}[(f(x) + \epsilon - \hat{f}(x)^2] \\ &= \mathbb{E}[(f(x) - \hat{f}(x))^2] + \mathbb{E}(\epsilon^2) + 2\mathbb{E}((f(x) - \hat{f}(x)\epsilon) \\ &= \mathbb{E}[(f(x) - \hat{f}(x))^2] + \sigma_\epsilon^2 + 2\mathbb{E}((f(x) - \hat{f}(x))\hat{E}(\epsilon)\end{aligned} \tag{219}$$

Further solving 219, we get,

$$\begin{aligned}\mathbb{E}[(f(x) - \hat{f}(x))^2] &= \mathbb{E}[(f(x) - \mathbb{E}[\hat{f}(x)]) - (\hat{f}(x) - \mathbb{E}[\hat{f}(x)]))^2] \\ &= \mathbb{E}[(\mathbb{E}[\hat{f}(x)] - (f(x)))^2] + \mathbb{E}[(\mathbb{E}[\hat{f}(x)] - (\hat{f}(x)))^2] \\ &\quad - 2\mathbb{E}[(\mathbb{E}[\hat{f}(x)] - (f(x)))(\mathbb{E}[\hat{f}(x)] - (\hat{f}(x)))] \\ &= bias(\hat{f}(x)^2) + var(\hat{f}(x)) + \epsilon\end{aligned} \tag{220}$$

The irreducible error cannot be reduced regardless of the algorithm used. It is the error introduced from the chosen framing of the problem and may be caused by factors like unknown variables that influence the mapping of the input variables to the output variable. The other two come under reducible error, where these can be altered. Bias and variance are two important concepts in statistics that are related to the accuracy and precision of statistical models or estimators.

3.6 Bias

Bias refers to the systematic error or tendency of a statistical estimator or model to consistently overestimate or underestimate the true population parameter or relationship between variables. Models with high bias are simple and fail to capture the complexity of the data. Hence, such models lead to higher training and testing errors. Low bias corresponds to a good fit to the training dataset.

$$bias[\hat{f}(x)] = E[\hat{f}(x)] - f(x)$$

- **Low Bias**: Suggests fewer assumptions about the form of the target function.
- **High Bias**: Suggests more assumptions about the form of the target function.

3.7 Variance

Variance refers to the variability or fluctuation in the estimated values of a statistical estimator or model due to random sampling or random error. A high variance estimator or model produces widely varying results across different samples, even if the average of these results is close to the true parameter value. High variance can result from overfitting a model to the sample data, using a complex model with many parameters, or insufficient sample size.

$$var(\hat{f}(x)) = E[(\hat{f}(x) - E[\hat{f}(x)])^2]$$

Any supervised machine learning algorithm aims to achieve low bias and low variance. In turn, the algorithm should achieve good prediction performance. You can see a general trend in the examples above:

- Linear machine learning algorithms often have a high bias but a low variance.
- Nonlinear machine learning algorithms often have a low bias but a high variance.

We cannot calculate the real bias and variance error terms because we do not know the underlying target function. Nevertheless, as a framework, bias, and variance provide the tools to understand the behavior of machine learning algorithms in the pursuit of predictive performance.

3.8 Confusion Matrix

In the field of machine learning, a confusion matrix is a table that is used to evaluate the performance of a classification model. It shows the number of true positives, false positives, true negatives, and false negatives produced by the model for a given data set. In a binary classification problem, where the

model is trained to distinguish between two classes, a confusion matrix would have four cells:

- **True Positives (TP)**: This refers to the cases where the model correctly predicted a positive outcome for positive input.
- **False Positives (FP)**: This refers to the cases where the model incorrectly predicted a positive outcome for negative input.
- **True Negatives (TN)**: This refers to the cases where the model correctly predicted a negative outcome for negative input.
- **False Negatives (FN)**: This refers to the cases where the model incorrectly predicted a negative outcome for positive input.

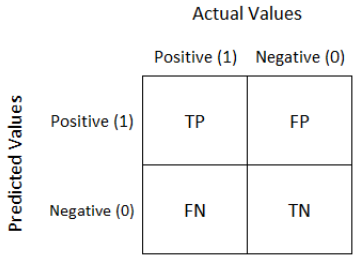


Figure 6: Confusion Matrix

3.9 Accuracy, Precision & Recall

Accuracy, precision, and recall are all metrics commonly used to evaluate the performance of a classification model based on the confusion matrix.

3.9.1 Accuracy

This metric measures the model's overall performance or how often it correctly predicts the input data class. It's calculated as the ratio of correct predictions (true positives and negatives) to the total number of predictions. The formula for accuracy is:

$$A = \frac{TP + TN}{TP + FP + TN + FN}$$

3.9.2 Precision

This metric measures the proportion of predicted positives that are actually true positives or the model's ability to avoid false positives. It's calculated as

the ratio of true positives to the total number of predicted positives. The formula for precision is:

$$P = \frac{TP}{TP + FP}$$

3.9.3 Recall

This metric measures the proportion of actual positives that are correctly identified by the model or the model's ability to avoid false negatives. It's calculated as the ratio of true positives to the total number of actual positives. The formula for the recall is:

$$R = \frac{TP}{TP + FN}$$

3.10 Statistical Inference

Statistical inference is the process of using data from a sample to make conclusions or predictions about a larger population. It involves analyzing and interpreting data through statistical methods to make inferences about the underlying characteristics of the population from which the sample was drawn. Statistical inference aims to make reliable and valid conclusions based on data from a sample that can be generalized to the larger population. However, statistical inference is subject to various sources of uncertainty and potential biases, such as sampling error, measurement error, and confounding variables, that can affect the accuracy and validity of the conclusions.

3.11 Law of Large Numbers

The law of large numbers is central to probability and statistics. It states that if you repeat an experiment independently many times and average the result, what you obtain should be close to the expected value. There are two main versions of the law of large numbers. They are called the weak and strong laws of large numbers. The difference between them is mostly theoretical. Mathematically, The Law of Large Numbers tells us that as the sample size increases and goes to ∞, the probability that $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i = \mu$ approaches 1.
Weak law of Large number states that Let $X_1, X_2, X_3...$ be the random variables with finite expected value $\mathrm{E}[X_i] = \mu < \infty$,
Then for any $\epsilon > 0$

$$\lim_{x \to \infty} P(|\overline{X} - \mu| \geq \epsilon) = 0$$

The proof of the weak law of large numbers is easier if we assume $\mathrm{Var(X)}=\epsilon^2$

is finite. In this case, we can use Chebyshev's inequality to write,

$$\begin{aligned} P(|\overline{X} - \mu|) &\leq \frac{Var(\bar{X})}{\epsilon^2} \\ &\leq \frac{Var(X)}{n\epsilon^2} \end{aligned} \tag{221}$$

is 0 as n goes to ∞

3.12 Central Limit Theorem

The central limit theorem (CLT) is one of the most important results in probability theory. It states that, under certain conditions, the sum of a large number of random variables is approximately normal. Here, we state a version of the CLT that applies to i.i.d. random variables. Suppose that $X_1, X_2, X_3, ...X_n$ are i.i.d. random variables with expected values $\mathbb{E}X_i = \mu < \infty$ and var(X_i)= $\sigma^2 < \infty$. Then the sample mean $\bar{X}$ has mean $\mathbb{E}\bar{X} = \mu$ and var($\bar{X}$) = $\frac{\sigma^2}{n}$, Thus normalized random variable

$$Z_n = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} = \frac{X_1 + X_2 + X_3 + ... + X_n - n\mu}{\sqrt{n}\sigma}$$

has mean $\mathbb{E}\bar{X} = 0$ and var($\bar{X}$) = 1 converges in distribution to the standard normal random variable as n goes to infinity, that is

$$\lim_{x \to \infty} p(Z_n \leq x) = \Phi(x)$$

for all x $\in \mathbb{R}$, where $\Phi(x)$ is the standard normal CDF.

4 Linear Algebra

4.1 Span

Theorem: Let $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_k, \omega \in \mathbf{V}$ and $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_k, \omega) = \mathbf{V}$, ω is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_k$. Then $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_k) = \mathbf{V}$.
Proof: To prove that $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_k) = \mathbf{V}$, we need to show that, every vector in vector space $\mathbf{V}$ can be written as a linear combination of vectors in the spanning set. Let $\mathbf{v} \in \mathbf{V}$ and as $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_k, \omega) = \mathbf{V}$, it can be written as linear combination of $\mathbf{v}_1, \mathbf{v}_2, ..., \mathbf{v}_k, \omega$,

$$\mathbf{v} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 \cdots + \alpha_k \mathbf{v}_k + \alpha_{k+1} \omega. \tag{222}$$

As ω is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_k$, we have

$$\mathbf{w} = \beta_1 \mathbf{v}_1 + \beta_2 \mathbf{v}_2 \ldots + \beta_k \mathbf{v}_k. \tag{223}$$

On substituting Equation 223 in Equation 222, we get

$$\begin{aligned} \mathbf{v} &= \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 \cdots + \alpha_k \mathbf{v}_k + \alpha_{k+1}(\beta_1 \mathbf{v}_1 + \beta_2 \mathbf{v}_2 \ldots + \beta_k \mathbf{v}_k). \\ &= (\alpha_1 + \beta_1 \alpha_{k+1})\, \mathbf{v}_1 + \cdots + (\alpha_k + \beta_k \alpha_{k+1})\, \mathbf{v}_k. \\ &= \text{span}(\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_k). \end{aligned}$$

Hence, $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_k) = \mathbf{V}$.

Proposition 1. *The vectors $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_k$ form basis of $\mathbf{V}$, iff $\forall \mathbf{v} \in \mathbf{V}$ can be written uniquely as $\mathbf{v} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 \cdots + \alpha_k \mathbf{v}_k$.*

Proof: Firstly, we prove that if $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_k$ form basis of $\mathbf{V}$, then $\forall \mathbf{v} \in \mathbf{V}$ can be written uniquely as $\mathbf{v} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 \cdots + \alpha_k \mathbf{v}_k$. This is proven by contradiction. Assume $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_k$ form basis of $\mathbf{V}$, Let $\mathbf{v} \in \mathbf{V}$, then

$$\mathbf{v} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 \cdots + \alpha_k \mathbf{v}_k. \tag{224}$$

Assume contrary that $\exists \beta_1, \cdots, \beta_k \in \mathbb{F}$, such that

$$\mathbf{v} = \beta_1 \mathbf{v}_1 + \beta_2 \mathbf{v}_2 \cdots + \beta_k \mathbf{v}_k. \tag{225}$$

On subtracting Equations 224, 225 we get

$$\mathbf{0}_v = (\alpha_1 - \beta_1)\mathbf{v}_1 + \cdots + (\alpha_k - \beta_k)\mathbf{v}_k.$$

Since, $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_k$ are linearly independent,

$$\begin{aligned} (\alpha_1 - \beta_1) = \cdots = (\alpha_k - \beta_k) = 0. \\ \alpha_i = \beta_i \quad \forall i \in [k]\,. \end{aligned}$$

Hence, $\forall \mathbf{v} \in \mathbf{V}$ can be written uniquely as $\mathbf{v} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 \cdots + \alpha_k \mathbf{v}_k$.

Now we prove the second part, assume $\mathbf{v} \in \mathbf{V}$ can be written uniquely as $\mathbf{v} = \alpha_1\mathbf{v}_1 + \alpha_2\mathbf{v}_2 \cdots + \alpha_k\mathbf{v}_k$. So, $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_k) = \mathbf{V}$. Consider,

$$\mathbf{0}_v = \alpha_1\mathbf{v}_1 + \cdots + \alpha_k\mathbf{v}_k.$$

But,

$$\mathbf{0}_v = 0.\mathbf{v}_1 + \cdots + 0.\mathbf{v}_k.$$

From uniqueness property, $\alpha_i = 0 \quad i \in [k]$. So, $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_k$ are linearly independent.Hence, $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_k$ form basis of $\mathbf{V}$.
Hence proved.

Proposition 2. *Let $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_r$ are linearly independent vectors in $\mathbf{V}$. Then we can extend the sequence to a basis $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_k$ where $k \geq r$.*

Proof: Let $\omega_1, \cdots, \omega_k$ be spanning set of $\mathbf{V}$. Consider $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_r, \omega_1, \cdots, \omega_k$ and apply elimination trick (eliminating linearly dependent vectors from the sequence). Since, $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_r$ are linearly independent, they won't get deleted in elimination. Therefore, $k \geq r$.

4.2 Linear Transformation

Definition 1. Let $\mathbf{U}$, $\mathbf{V}$ be two vector space over same field $\mathbb{F}$. A linear transformation $T : \mathbf{U} \to \mathbf{V}$ is a map/function such that

1. $T(\mathbf{u}_1 + \mathbf{u}_2) = T(\mathbf{u}_1) + T(\mathbf{u}_2) \quad \forall \mathbf{u}_1, \mathbf{u}_2 \in \mathbf{U}$.
2. $T(\alpha\mathbf{u}) = \alpha T(\mathbf{u}) \quad \forall \alpha \in \mathbb{F} \quad \mathbf{u} \in \mathbf{U}$.

In other words,

$$T(\alpha\mathbf{u}_1 + \beta\mathbf{u}_2) = \alpha T(\mathbf{u}_1) + \beta T(\mathbf{u}_2) \quad \forall \alpha, \beta \in \mathbb{F} \quad \mathbf{u}_1, \mathbf{u}_2 \in \mathbf{U}.$$

Let's consider an example of a linear transformation,

Example 1. $\mathbf{U} = \mathbb{R}^3, \mathbf{V} = \mathbb{R}^2$. Define $T : \mathbf{U} \to \mathbf{V}$ such that $T(u_1, u_2, u_3) = (u_1, u_2)$. Show T is a linear transformation

We prove that T is a linear transformation by showing that

$$T(\alpha\mathbf{u}_1 + \beta\mathbf{u}_2) = \alpha T(\mathbf{u}_1) + \beta T(\mathbf{u}_2) \quad \forall \alpha, \beta \in \mathbb{F} \quad \mathbf{u}_1, \mathbf{u}_2 \in \mathbf{U}.$$

Let $\mathbf{x} = (x_1, x_2, x_3), \mathbf{y} = (y_1, y_2, y_3) \in \mathbf{U}$. Then

$$T(x_1, x_2, x_3) = (x_1, x_2).$$
$$T(y_1, y_2, y_3) = (y_1, y_2).$$

$$\begin{aligned} T(\alpha\mathbf{x}+\beta\mathbf{y}) &= T\left[\alpha\left(x_1,x_2,x_3\right)+\beta\left(y_1,y_2,y_3\right)\right]. \\ &= T\left[\left(\alpha x_1+\beta y_1,\alpha x_2+\beta y_2,\alpha x_3+\beta y_3\right)\right]. \\ &= \alpha\left(x_1,x_2\right)+\beta\left(y_1,y_2\right). \\ &= \alpha T\left(x_1,x_2,x_3\right)+\beta T\left(y_1,y_2,y_3\right). \\ &= \alpha T(\mathbf{x})+\beta T(\mathbf{y}). \end{aligned}$$

Hence, T is a linear transformation.

Lemma 1. *If $T:\mathbf{U}\to\mathbf{V}$ be a linear transformation. Then*

1. $T\left(\mathbf{0}_u\right)=\mathbf{0}_v$.
2. $T(-\mathbf{u})=-T(\mathbf{u}) \quad \forall\mathbf{u}\in\mathbf{U}$.

Proof: 1. Consider,

$$\begin{aligned} \mathbf{0}_u &= \mathbf{0}_u+\mathbf{0}_u. \\ T\left(\mathbf{0}_u\right) &= T(\mathbf{0}_u+\mathbf{0}_u). \\ &= T(\mathbf{0}_u)+T(\mathbf{0}_u). \\ &= 2\times T(\mathbf{0}_u). \\ T(\mathbf{0}_u) &= \mathbf{0}_v. \end{aligned}$$

2. Proof of second part is obvious by taking $\alpha=1$ in $T(\alpha\mathbf{u})=\alpha T(\mathbf{u})$ of Definition 1

Definition 2. The image of a linear transformation is the set of all vectors it can produce on doing a linear transformation of vectors in a given vector space. Let $T:\mathbf{U}\to\mathbf{V}$ be a linear transformation. Then, image of T,

$$\mathrm{Im}(T)=\{T(\mathbf{u})\mid\mathbf{u}\in\mathbf{U}\}.$$

Definition 3. The kernel of a linear transformation is the set of all vectors that on linear transformation are mapped to the null vector. Let $T:\mathbf{U}\to\mathbf{V}$ be a linear transformation. Then, the kernel of T,

$$\mathrm{Ker}(T)=\{\mathbf{u}\in\mathbf{U}\mid T(\mathbf{u})=\mathbf{0}_v\}.$$

Proposition 3. *Let $T:\mathbf{U}\to\mathbf{V}$ be a linear transformation. Then,*

1. $\mathrm{Im}(T)$ *is a subspace of* $\mathbf{V}$.
2. $\mathrm{Ker}(T)$ *is a subspace of* $\mathbf{U}$.

Proof:

1. Let $\mathbf{v}_1,\mathbf{v}_2\in\mathrm{Im}(T)$. Then $\exists\mathbf{u}_1,\mathbf{u}_2\in\mathbf{U}$ such that

$$\begin{aligned} T(\mathbf{u}_1) &= \mathbf{v}_1. \\ T(\mathbf{u}_2) &= \mathbf{v}_2. \end{aligned}$$

We need to show that $(\alpha\mathbf{v}_1 + \beta\mathbf{v}_2) \in \text{Im}(T)$.

$$\begin{aligned}\alpha\mathbf{v}_1 + \beta\mathbf{v}_2. &= \alpha T(\mathbf{u}_1) + \beta T(\mathbf{u}_2).\\ &= T(\alpha\mathbf{u}_1 + \beta\mathbf{u}_2) \in \text{Im}(T).\end{aligned}$$

Hence, $\text{Im}(T)$ is a subspace of $\mathbf{V}$.

2. Let $\mathbf{u}_1, \mathbf{u}_2 \in \text{Ker}(T)$. Then

$$\begin{aligned}T(\mathbf{u}_1) &= \mathbf{0}_v.\\ T(\mathbf{u}_2) &= \mathbf{0}_v.\end{aligned}$$

We need to show that $(\alpha\mathbf{v}_1 + \beta\mathbf{v}_2) \in \text{Ker}(T)$.

$$\begin{aligned}T(\alpha\mathbf{u}_1 + \beta\mathbf{u}_2) &= \alpha T(\mathbf{u}_1) + \beta T(\mathbf{u}_2).\\ &= \alpha\mathbf{0}_v + \beta\mathbf{0}_v.\\ &= \mathbf{0}_v.\end{aligned}$$

So, $(\alpha\mathbf{v}_1 + \beta\mathbf{v}_2) \in \text{Ker}(T)$. Hence, $\text{Ker}(T)$ is a subspace of $\mathbf{U}$.

Example 2. Let $\mathbf{U} = \mathbf{V} = \mathbb{R}^2$. Define $T : \mathbf{U} \to \mathbf{V}$ such that $T(\mathbf{v})$ gives the vector obtained by rotating $\mathbf{v}$ by an θ in the anti-clockwise direction.

To find, $T(\mathbf{v})$, we take the component of $\mathbf{v}$ along the x and y axes, and then find the linear transformation of the components separately (which is easier to find).
Consider $(\alpha, \beta) \in \mathbb{R}^2$,

Figure 7: Rotation by θ in anti-clockwise direction

$$\begin{aligned}(\alpha, \beta) &= \alpha(1, 0) + \beta(0, 1).\\ T(\alpha, \beta) &= T\left[\alpha(1, 0) + \beta(0, 1)\right].\\ &= \alpha T(1, 0) + \beta T(0, 1).\\ &= \alpha(\cos\theta, \sin\theta) + \beta(-\sin\theta, \cos\theta).\\ &= (\alpha\cos\theta - \beta\sin\theta, \alpha\sin\theta + \beta\cos\theta).\end{aligned}$$

The following section contains the definitions needed to prove the Rank Nullity Theorem. For the purpose of understanding this theorem and its proof, we must first be familiar with the concepts of Kernel and Image of a Linear Transformation.

4.3 Rank and Nullity

Definition 4. Dimension of a Vector Space: Dimension of a vector space $\mathcal{V}$ is given by the number of vectors (n) in its basis $\dim(\mathcal{V}) = n$.

Definition 5. Rank and Nullity of a Linear Transformation: Let $\mathbf{T} : \mathcal{U} \to \mathcal{V}$ be a linear map

Then the dimension of Image space of $\mathbf{T}$ is defined as its Rank, $\dim(\text{Im}(\mathbf{T})) = \text{rank}(\mathbf{T})$.

Then the dimension of Kernel space of $\mathbf{T}$ is defined as its Nullity, $\dim(\text{Ker}(\mathbf{T})) = \text{null}(\mathbf{T})$.

4.4 Rank-Nullity Theorem

The following theorem states that the dimension of the vector space on which linear transformation is applied can be determined by the sum of the rank and nullity of that transformation. The following is the mathematical statement for the theorem and its proof. **Theorem:** Let $\mathcal{U}$ and $\mathcal{V}$ be two vector spaces over $\mathbb{F}$. Let $\mathbf{T} : \mathcal{U} \to \mathcal{V}$ be a linear transformation then

$$\text{rank}(\mathbf{T}) + \text{null}(\mathbf{T}) = \dim(\mathcal{U}).$$

Proof: Since $\mathcal{U}$ is finite dimensional, $\text{Ker}(\mathbf{T})$ is subspace of $\mathcal{U}$ is also finite dimensional. Let

$$\dim(\text{Ker}(\mathbf{T})) = \text{null}(\mathbf{T}) = s.$$

Let $\mathbf{e_1}, \mathbf{e_2} \dots \mathbf{e_s}$ are basis vectors of $Ker(\mathbf{T})$. We can extend $\mathbf{e_1} \dots \mathbf{e_s}$ to basis vector of $\mathcal{U}$. Let

$\mathbf{e_1}, \mathbf{e_2} \dots \mathbf{e_s}, \mathbf{f_1}, \mathbf{f_2} \dots \mathbf{f_r}$ are basis vectors of $\mathbf{U}$.
$\dim(\mathcal{U}) = r + s$.

We need to prove $\text{rank}(\mathbf{T}) = r$, Consider Linear Transform on basis of $\mathbf{U}$

$$\mathbf{T}(\mathbf{e_1}), \mathbf{T}(\mathbf{e_2}) \dots \mathbf{T}(\mathbf{e_s}), \mathbf{T}(\mathbf{f_1}), \mathbf{T}(\mathbf{f_2}) \dots \mathbf{T}(\mathbf{f_r}) \text{ should span } \text{Im}(\mathbf{T}).$$

Since basis can represent every vector in $\mathcal{U}$ its linear transformation has to represent every vector in Image space $\mathcal{U}$. We know that for some vector $\mathbf{k} \in \text{Ker}(\mathbf{T})$, $\mathbf{T}(\mathbf{k}) = \mathbf{0_V}$.

$$\mathbf{T}(\mathbf{e_1}) = \dots = \mathbf{T}(\mathbf{e_s}) = \mathbf{0_V} \ (\because \mathbf{e_1} \dots \mathbf{e_s} \in \text{Ker}(\mathbf{T})).$$
$$\implies \text{span}(\mathbf{T}(\mathbf{f_1}), \mathbf{T}(\mathbf{f_2}) \dots \mathbf{T}(\mathbf{f_r})) = \text{Im}(\mathbf{T}).$$

We need to show $\mathbf{T}(\mathbf{f_1}) \dots \mathbf{T}(\mathbf{f_r})$ form basis of image space of $\mathbf{T}$ so that $\text{rank}(\mathbf{T}) = r$. as we have already shown it spans $\text{Im}(\mathbf{T})$ we need to show $\mathbf{T}(\mathbf{f_1}) \dots \mathbf{T}(\mathbf{f_r})$ are linearly independent.

Assume $\sum_{i=1}^{r} \alpha_i \mathbf{T}(\mathbf{f_i}) = \mathbf{0_V}$ for some $\alpha_i \in \mathbb{R}$, $\forall i \in \{1, 2, \ldots, r\}$. To prove they are linearly independent we need to show $\alpha_i = 0$, $\forall i \in \{1, 2, \ldots, r\}$.

$\alpha_1 \mathbf{T}(\mathbf{f_1}) + \alpha_2 \mathbf{T}(\mathbf{f_2}) + \cdots + \alpha_r \mathbf{T}(\mathbf{f_r}) = \mathbf{0_V}$.
$\mathbf{T}(\alpha_1 \mathbf{f_1} + \cdots + \alpha_r \mathbf{f_r}) = \mathbf{0_V}$.
$\implies \alpha_1 \mathbf{f_1} + \cdots + \alpha_r \mathbf{f_r} \in \text{Ker}(\mathbf{T})$.
$\alpha_1 \mathbf{f_1} + \cdots + \alpha_r \mathbf{f_r} = \beta_1 \mathbf{e_1} + \cdots + \beta_s \mathbf{e_s}$ ($\because \mathbf{e_1} \ldots \mathbf{e_s}$ form basis of $\text{Ker}(\mathbf{T})$).
$\alpha_1 \mathbf{f_1} + \ldots \alpha_r \mathbf{f_r} - \beta_1 \mathbf{e_1} - \cdots - \beta_s \mathbf{e_s} = \mathbf{0_V}$.
$\because \mathbf{e_1}, \mathbf{e_2} \ldots \mathbf{e_s}, \mathbf{f_1}, \mathbf{f_2} \ldots \mathbf{f_r}$ are linearly independent as they form basis for $\mathcal{U}$.
$\implies \alpha_1 = \alpha_2 = \ldots \alpha_r = \beta_1 = \beta_2 = \ldots \beta_s = 0$.
$\therefore \mathbf{T}(\mathbf{f_1}) \ldots \mathbf{T}(\mathbf{f_r})$ are linearly independent.

Hence, from above equations $\mathbf{T}(\mathbf{f_1}) \ldots \mathbf{T}(\mathbf{f_r})$ form basis of $\text{Im}(\mathbf{T})$.
$\dim(\text{Im}(\mathbf{T})) = \text{null}(\mathbf{T}) = r$.

$$\therefore \text{rank}(\mathbf{T}) + \text{null}(\mathbf{T}) = \dim(\mathbf{U}).$$

4.5 QR decomposition

4.5.1 Linear Least Squares Problem

As discussed in the previous lecture, we had taken a situation where we have been given certain *features* of a house and the price of the corresponding house. Our task is to fit a linear equation curve that best represents this data. Our equation would look something like this.

$$\mathbf{Ax} \approx \mathbf{b}, \tag{226}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \ldots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \ldots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \ldots & a_{mn} \end{bmatrix}_{m \times n} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1} \approx \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}_{m \times 1}. \tag{227}$$

Here $\mathbf{A}$ is the matrix where each row represents the feature vector of that particular house, and $\mathbf{b}$ is the ground truth price of the house. Our task is to find an $\mathbf{x}$ that best represents this equation.
From the above equation, we can see that, for any i^{th} house, its corresponding price would be.

$$\begin{bmatrix} a_{i1} & a_{i2} & \ldots a_{in} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = (\mathbf{Ax})_i = \hat{b}_i. \tag{228}$$

Here $\hat{b}_i$ represents the predicted price of any i^{th} house for a certain $\mathbf{x}$.
Below is the objective function that we need to optimize.

$$\mathcal{L} = \min_{\mathbf{x}\in\mathbb{R}^n} \|\mathbf{b} - \hat{\mathbf{b}}\|^2 = \min_{\mathbf{x}\in\mathbb{R}^n} \|\mathbf{b} - \mathbf{A}\mathbf{x}\|^2. \tag{229}$$

Here $\mathcal{L}$ is the loss we are trying to minimize. The reason we have taken $\|.\|^2$ is that we trying to find the least square problem solution. Now let us try to find the optimal $\mathbf{x}$.

$$\|\mathbf{b} - \mathbf{A}\mathbf{x}\|^2 = (\mathbf{A}\mathbf{x} - \mathbf{b})^T(\mathbf{A}\mathbf{x} - \mathbf{b}) \tag{230}$$
$$= \mathbf{x}^T\mathbf{A}^T\mathbf{A}\mathbf{x} - 2\mathbf{b}^T\mathbf{A}\mathbf{x} + \mathbf{b}^T\mathbf{b} \tag{231}$$
$$= \mathbf{x}^T\mathbf{H}\mathbf{x} - 2\mathbf{b}^T\mathbf{A}\mathbf{x} + \mathbf{b}^T\mathbf{b}. \tag{232}$$

Let us take the derivative of $\mathcal{L}$ with respect to $\mathbf{x}$.

$$\frac{\partial}{\partial\mathbf{x}}\,\mathbf{x}^T\mathbf{H}\mathbf{x} = 2\mathbf{H}\mathbf{x}, \tag{233}$$
$$\frac{\partial}{\partial\mathbf{x}}\,\mathbf{b}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b}, \tag{234}$$
$$\frac{\partial}{\partial\mathbf{x}}\,\mathcal{L} = 2\mathbf{H}\mathbf{x} - 2\mathbf{A}^T\mathbf{b}. \tag{235}$$

Putting $\frac{\partial}{\partial\mathbf{x}}\,\mathcal{L}$ as 0, we get.

$$2\mathbf{H}\mathbf{x} - 2\mathbf{A}^T\mathbf{b} = 0 \tag{236}$$
$$2\mathbf{A}^T\mathbf{A}\mathbf{x} - 2\mathbf{A}^T\mathbf{b} = 0 \tag{237}$$
$$\mathbf{A}^T\mathbf{A}\mathbf{x} = \mathbf{A}^T\mathbf{b} \tag{238}$$
$$\therefore \boxed{\mathbf{x} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{b}}. \tag{239}$$

The value of $\mathbf{x}$ that we got here looks similar to a linear transformation that we have studied before. It is the orthogonal projection.
$\mathbf{x}$ is the orthogonal projection of $\mathbf{b}$ onto the subspace spanned by columns of $\mathbf{A}$. We will now see how QR decomposition can help us find the projection.

4.5.2 QR Decomposition

Definition 6. Every matrix $\mathbf{A}_{m\times n}$ can be expressed as a product of $\mathbf{Q}_{m\times m}$ and $\mathbf{R}_{m\times n}$, assuming $m \geq n$. Here $\mathbf{Q}$ is an **Orthogonal Matrix** and $\mathbf{R}$ is an **Upper Triangular Matrix**.

Since $\mathbf{R}$ is a thin matrix, then the bottom $(m-n)$ rows will be 0, similarly, $\mathbf{Q}$ can also be partitioned in a way to be multiplied by $\mathbf{R}$. The decomposition will look like this.

$$\mathbf{A} = \mathbf{Q}\mathbf{R} \tag{240}$$
$$= \begin{bmatrix}\mathbf{Q}_{1_{m\times n}} & \mathbf{Q}_{2_{m\times(m-n)}}\end{bmatrix}\begin{bmatrix}\mathbf{R}_{1_{n\times n}} \\ \mathbf{0}_{(m-n)\times n}\end{bmatrix} \tag{241}$$
$$= \mathbf{Q}_1\mathbf{R}_1. \tag{242}$$

$\mathbf{Q}_1$ and $\mathbf{Q}_2$ both have orthogonal columns, and $\mathbf{R}_1$ is an upper triangular square matrix. We will now substitute Eq. 240 into Eq. 239.

$$\begin{aligned}
\mathbf{x} &= (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{b} && (243)\\
&= ((\mathbf{QR})^T\mathbf{QR})^{-1}(\mathbf{QR})^T\mathbf{b} && (244)\\
&= \mathbf{QQ}^T\mathbf{b} && (245)\\
&= \Pi_{\mathbf{Q}}(\mathbf{b}). && (246)
\end{aligned}$$

Here $\Pi_{\mathbf{Q}}$ denotes the orthogonal projection of $\mathbf{b}$ onto the subspace spanned by columns of $\mathbf{Q}$.

4.5.3 Gram Schmidt Orthogonalization Process

We have studied till now about the application of **QR** decomposition to solve linear least squares problems, but it is also useful in calculating the *eigenvectors* and *eigenvalues* of matrices as well.
In this section, we will study how to find the **QR** decomposition of a matrix **Q**. The method (as seen from the title) is called Gram-Schmidt Orthogonalization Process.
The decomposition is as we saw before:

$$\mathbf{A}_{m\times n} = \mathbf{Q}_{m\times m}\mathbf{R}_{m\times n}. \tag{247}$$

Let us take as following.

$$\mathbf{A} = \begin{bmatrix}\mathbf{a}_{:1} & \mathbf{a}_{:2} & \dots & \mathbf{a}_{:n}\end{bmatrix}, \tag{248}$$

$$\mathbf{Q} = \begin{bmatrix}\mathbf{e}_{:1} & \mathbf{e}_{:2} & \dots & \mathbf{e}_{:m}\end{bmatrix}. \tag{249}$$

The process is as follows.

$$\mathbf{u_1} = \mathbf{a}_{:1}, \qquad \mathbf{e}_{:1} = \frac{\mathbf{u_1}}{\|\mathbf{u_1}\|}, \tag{250}$$

$$\mathbf{u_2} = \mathbf{a}_{:2} - \langle \mathbf{a}_{:2}, \mathbf{e}_{:1}\rangle \mathbf{e}_{:1}, \qquad \mathbf{e}_{:2} = \frac{\mathbf{u_2}}{\|\mathbf{u_2}\|}, \tag{251}$$

$$\mathbf{u_3} = \mathbf{a}_{:3} - \langle \mathbf{a}_{:3}, \mathbf{e}_{:1}\rangle \mathbf{e}_{:1} - \langle \mathbf{a}_{:3}, \mathbf{e}_{:2}\rangle \mathbf{e}_{:2}, \qquad \mathbf{e}_{:3} = \frac{\mathbf{u_3}}{\|\mathbf{u_3}\|}, \tag{252}$$

$$\therefore \mathbf{u_{k+1}} = \mathbf{a}_{:k+1} - \sum_{i=1}^{k} \langle \mathbf{a}_{:k+1}, \mathbf{e}_{:i}\rangle, \qquad \mathbf{e}_{:k+1} = \frac{\mathbf{u_{k+1}}}{\|\mathbf{u_{k+1}}\|}. \tag{253}$$

$$\tag{254}$$

Hence, we get the values of **Q**, now for **R**, we need to write $\mathbf{a}_{:i}$ as a linear

combination of $\mathbf{e_{:j}}$.

$$\mathbf{a_{:1}} = \langle \mathbf{a_{:1}}, \mathbf{e_{:1}} \rangle \mathbf{e_{:1}}, \tag{255}$$

$$\mathbf{a_{:2}} = \langle \mathbf{a_{:2}}, \mathbf{e_{:1}} \rangle \mathbf{e_{:1}} + \langle \mathbf{a_{:2}}, \mathbf{e_{:2}} \rangle \mathbf{e_{:2}}, \tag{256}$$

$$\mathbf{a_{:3}} = \langle \mathbf{a_{:3}}, \mathbf{e_{:1}} \rangle \mathbf{e_{:1}} + \langle \mathbf{a_{:3}}, \mathbf{e_{:2}} \rangle \mathbf{e_{:2}} + \langle \mathbf{a_{:3}}, \mathbf{e_{:3}} \rangle \mathbf{e_{:3}}, \tag{257}$$

$$\therefore \mathbf{a_{:i}} = \sum_{j=1}^{i} \langle \mathbf{a_{:i}}, \mathbf{e_{:j}} \rangle \mathbf{e_{:j}}. \tag{258}$$

$\mathbf{R}$ matrix will consist of ith column as the coefficients of each of $\mathbf{e_{:j}}$ that sum up to $\mathbf{a_{:i}}$.
Let us solve an example

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \tag{259}$$

$$\mathbf{u_1} = \mathbf{a_{:1}} = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \qquad \mathbf{e_{:1}} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}, \tag{260}$$

$$\mathbf{u_2} = \mathbf{a_{:2}} - \frac{1}{\sqrt{2}} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{\sqrt{2}} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}, \qquad \mathbf{e_{:2}} = \begin{pmatrix} \frac{1}{\sqrt{6}} \\ \frac{-1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{pmatrix}, \tag{261}$$

$$\mathbf{u_3} = \mathbf{a_{:3}} - \frac{1}{\sqrt{2}} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix} - \frac{1}{\sqrt{6}} \begin{pmatrix} \frac{1}{\sqrt{6}} \\ \frac{-1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{pmatrix}, \tag{262}$$

$$= \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} - \frac{1}{\sqrt{2}} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix} - \frac{1}{\sqrt{6}} \begin{pmatrix} \frac{1}{\sqrt{6}} \\ \frac{-1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{pmatrix}, \qquad \mathbf{e_{:3}} = \begin{pmatrix} \frac{-1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{pmatrix}, \tag{263}$$

$$\mathbf{Q} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{-1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{-1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}, \tag{264}$$

$$\mathbf{R} = \begin{bmatrix} \sqrt{2} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \frac{3}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{2}{\sqrt{3}} \end{bmatrix}. \tag{265}$$

Claim 1. *All the vectors of* $\mathbf{Q}$ *matrix i.e.* $\mathbf{e_{:1}}, \mathbf{e_{:2}}, \ldots, \mathbf{e_{:n}}$ *are orthonormal.*

Proof: We can say that each of $\mathbf{e_{:i}}$ was constructed such that from each $\mathbf{a_{:i}}$, the component of $\mathbf{e_{:i-1}}$ was removed, and hence, by construction, we can say that all vectors are orthonormal.

Claim 2. *span*$(\mathbf{e_{:1}}, \mathbf{e_{:2}}, \ldots, \mathbf{e_{:n}})$ = *span*$(\mathbf{a_{:1}}, \mathbf{a_{:2}}, \ldots, \mathbf{a_{:n}})$.

Proof: We have already seen that $\mathbf{a_{:i}}$ can be written as a linear combination of $\mathbf{e_{:j}}$, hence any linear combination of columns of $\mathbf{Q}$ can be written as linear combination of $\mathbf{e_{:j}}$.

4.6 Singular Value Decomposition

4.6.1 Matrix Norms

The definition of a matrix norm is analogous to the definition of a right singular vector. Here is the definition.

Definition 7. The value of the $p-norm$ of a matrix A is denoted and defined as follows.

$$\|\mathbf{A}\|_p = \max_{\|\mathbf{v}\|=1} \|\mathbf{Av}\|_p.$$

Definition 8. The *Frobenius Norm* of a Matrix A is denoted by $\|A\|_F$ is

$$\|\mathbf{A}\|_F^2 = \sum_i \sum_j |a_{ij}|^2.$$

4.6.2 Singular Vectors

4.6.3 Intuition

In case we ever wanted to find vectors that actually maximize $\|\mathbf{Av}\|$, it would be clear that $\mathbf{v}$ surely should stay "away" from the Nullspace of A. "Away" because for a fixed norm of $\|\mathbf{v}\| = 1$, $\mathbf{v}$ should not have any components belonging to the Nullspace of A (otherwise you can see that some part $\mathbf{v}$ would go waste in order to maximize $\|\mathbf{Av}\|$). The Space of vectors that are perpendicular to the Nullspace of A is the Rowspace. So Intuitively, we have established that choosing $v \in \text{Col}(\mathbf{A}^T)$ is a good choice.

4.6.4 Definitions

We define the right singular vectors, left singular vectors and singular values of $\mathbf{A}$ as follows. The $i'th$ right singular vector of A is $\mathbf{v_i}$ where

$$\mathbf{v_i} = \underset{\mathbf{v} \perp \mathbf{v_j} \forall j<i, \|\mathbf{v}\|=1}{\arg\max} \|\mathbf{Av}\|.$$

The definition of the $i'th$ left singular vector(u_i) and the $i'th$ singular value (σ_i) is as follows

$$\mathbf{Av_i} = \sigma_i \mathbf{u_i} \ , \|\mathbf{u_i}\| = \|\mathbf{v_i}\| = 1.$$

4.6.5 Properties of the Singular Vectors

Firstly, it is obvious that the **right singular vectors are orthogonal to each other.** This is because of the definition. A question would be to ask exactly how many such orthogonal vectors $\mathbf{v_i}$ actually exist for which the

singular value $\sigma_i \neq 0$ (Since there is only an r dimensional row space, of which we are exhausting one vector on each step, it would make sense that the number of right singular vectors that we would get for which $Av_i \neq 0$ will be only $r = \text{rank}(A)$.). Let us prove this more rigorously.

4.7 Rank Decomposition

Theorem: A, a matrix of rank r can be written in the following format in terms of its left and right singular vectors

$$\mathbf{A} = \sum_{i=1}^{rank(\mathbf{A})} \sigma_i \mathbf{u_i} \mathbf{v_i^T}.$$

Proof: Consider $B_1 = A - \sigma_1 u_1 v_1^T$, let us analyze the right singular vectors of this new matrix B_1. Let us then evaluate

$$\mathbf{B_1 v} = \mathbf{A v} - \sigma_1 \mathbf{u_1} \mathbf{v_1^T} \mathbf{v}.$$

Then we take the maximization of the norm of Bv constrained on the perpendicularity with v_1, the first right singular vector of A.

$$\underset{v \perp v_1, \|v\|=1}{\arg\max} \|\mathbf{B_1 v}\| = \underset{v \perp v_1, \|v\|=1}{\arg\max} \|\mathbf{A v} - \sigma_1 \mathbf{u_1} \mathbf{v_1^T} \mathbf{v}\| = \underset{v \perp v_1, \|v\|=1}{\arg\max} \|\mathbf{A v}\| = \mathbf{v_2} \quad (\because v \perp v_1).$$

Now comes, the critical argument, we have to prove that the LHS is actually the first singular vector of $\mathbf{B_1}$. i.e we need to prove that this form of constraint on v will not make a difference on the minimization of $\|\mathbf{B_1 v}\| \quad \forall v \in \mathbb{R}^n$.

Lemma 2.

$$\underset{v \perp v_1, \|v\|=1}{\arg\max} \|\mathbf{B_1 v}\|, \|\mathbf{v}\| = 1 = \underset{v \in \mathbb{R}^n, \|v\|=1}{\arg\max} \|\mathbf{B_1 v}\|.$$

where v_1 is the first singular vector of $\mathbf{A} = \mathbf{B} + \sigma_1 \mathbf{u_1} \mathbf{v_1^T}$.

Proof: We must show that for any v that maximizes the RHS, $v \perp v_1$
Assume the contrary, then we can construct a new vector v_n that will gives us a higher $\|\mathbf{B v}\|$.
Consider the vector

$$\mathbf{v_n} = \frac{\mathbf{v} - \langle \mathbf{v}, \mathbf{v_1} \rangle \mathbf{v_1}}{\|\mathbf{v} - \langle \mathbf{v}, \mathbf{v_1} \rangle \mathbf{v_1}\|}.$$

Now evaluate $\|\mathbf{B v_n}\|$ while remembering that $\mathbf{B v_1} = 0$

$$\|\mathbf{B v_n}\| = \frac{\|\mathbf{B v}\|}{\|\mathbf{v} - \langle \mathbf{v}, \mathbf{v_1} \rangle \mathbf{v_1}\|} \geq \|\mathbf{B v}\|.$$

The Denominator is lesser than one, since it is the length of the perpendicular component of v on v_1 Since this is a contradiction to v being the argument maximizer, we get that $v_{opt} \perp v_1$.

Now, that the Lemma is established, it is clear that the first singular vector of $\mathbf{B_1}$ is the second singular vector of $\mathbf{A}$. Now, we repeat this process so that we can generate the first r singular vectors of $\mathbf{A}$. i.e. we can get a $\mathbf{B_r}$, that is supposed to correspond to the value :-

$$\mathbf{v_1}(\mathbf{B_r}) = \mathbf{v_2}(\mathbf{B_{r-1}}) = \dots \mathbf{v_r}(\mathbf{B_1}) = \mathbf{v_{r+1}}(\mathbf{A}) = \underset{v \perp v_i \forall i \leq r}{\arg\max} \|\mathbf{Av}\| = 0.$$

From this, it follows that the first singular vector of the matrix B_r is zero! The only way this is possible is if the matrix $B_r = 0$!. (because the Nullspace of B_r is the entire space, implying it has zero rank).

$$\boxed{\mathbf{B_r} = \mathbf{A} - \sum_{\mathbf{i}}^{\mathbf{r}} \sigma_\mathbf{i} \mathbf{u_i} \mathbf{v_i^T} = \mathbf{0} \implies \mathbf{A} = \sum_{\mathbf{i}}^{\mathbf{r}} \sigma_\mathbf{i} \mathbf{u_i} \mathbf{v_i^T}}.$$

Another way to represent matrix A would be to write

$$\mathbf{A} = \mathbf{U_1} \mathbf{\Sigma_1} \mathbf{V_1^T},$$

where $\mathbf{U_1}$ = column arranged $\mathbf{u_i} \in \mathbb{R}^{m \times r}$
$\mathbf{V_1^T}$ = row arranged $\mathbf{v_i} \in \mathbb{R}^{r \times n}$
$\mathbf{\Sigma_1}$ = diagonal arranged $\sigma_i \in \mathbb{R}^{r \times r}$
Clearly, the matrix is now a sum of r independent rank-one Matrices. This means that the rank of the matrix is r.

4.7.1 Proving U_1 is semi-unitary

We proceed by contradiction, we will assume that $\mathbf{u_1}$ and $\mathbf{u_i}$ are not perpendicular to each other and then we will use this to construct a new vector $\mathbf{v}$ that maximizes $\|\mathbf{Av}\|$ better than $\mathbf{v_1}$.
We can use the vector

$$\mathbf{v} = \mathbf{v_1} + \epsilon \mathbf{v_i} \quad |\epsilon \geq 0$$

This vector when multiplied with A will give a higher value of $\|Av\|$when $u_1^T u_i \geq 0$.

$$\|\mathbf{Av}\| = \frac{\|\sigma_1 \mathbf{u_1} + \epsilon \sigma_i \mathbf{u_i}\|}{\sqrt{1+\epsilon^2}}, \tag{266}$$

$$\|\mathbf{Av}\|^2 = \frac{\sigma_1^2 + \epsilon^2 \sigma_i^2 + 2\epsilon \sigma_1 \sigma_i C}{1+\epsilon^2}. \tag{267}$$

If we can prove that for an appropriate choice of ϵ The value of $\epsilon^2 \sigma_i^2 + 2\epsilon \sigma_1 \sigma_i C \quad \forall C \geq 0$ will exceed $\epsilon^2 \sigma_1^2$ Our proof is complete,

$$\epsilon^2 \sigma_i^2 + 2\epsilon \sigma_1 \sigma_i C \geq \epsilon^2 \sigma_1^2,$$

$$\epsilon^2 (\sigma_1^2 - \sigma_i^2) - 2\epsilon \sigma_1 \sigma_i C \leq 0,$$

$$(\epsilon - 0)(\epsilon(\sigma_1^2 - \sigma_i^2) - 2\sigma_1\sigma_i C) \leq 0,$$

$$\epsilon \geq 0 \quad \& \quad \epsilon \leq \frac{2\sigma_1\sigma_i C}{\sigma_1^2 - \sigma_i^2}.$$

Since C is positive and $\sigma_1 \geq \sigma_i$, this value of ϵ is achievable if $\epsilon \geq 0$. This proves that $\mathbf{u_1} \perp \mathbf{u_i}$.

4.8 Full and thin SVD

Now that we have established that U_1 and V_1 are both unitary, we obtain that

$$\mathbf{A} = \mathbf{U_1}\mathbf{\Sigma_1}\mathbf{V_1^T}.$$

Now note that since all of these matrices are semi-unitary,

$$\text{Col}(\mathbf{A}) = \text{Col}(\mathbf{U_1}),$$

$$\text{Col}(\mathbf{A^T}) = \text{Col}(\mathbf{V_1}).$$

It would be nice top extend the bases that are described in an orthonormal way to get a full SVD decomposition, that will gives us the bases for the orthogonal complements of the above subspaces i.e. Left Nullspace and the Nullspace of the matrix A.

$$\boxed{\mathbf{A} = [\mathbf{U_1} \quad \mathbf{U_2}]_{m\times m} \begin{bmatrix} \mathbf{\Sigma_1} & 0 \\ 0 & 0 \end{bmatrix}_{m\times n} \begin{bmatrix} \mathbf{V_1^T} \\ \mathbf{V_2^T} \end{bmatrix}_{n\times n}}.$$

Note that $\mathbf{U}$ and $\mathbf{V}$ are now **Unitary** matrices. The extra results are:-

$$\text{Nul}(A^T) = \text{Col}(U_2),$$
$$\text{Nul}(A) = \text{Col}(V_2).$$

4.8.1 Additional Interpretations

4.8.2 Expressing the Frobenius Norm

Let $\mathbf{a_j}$ represent the row vectors of $\mathbf{A}$.

$$\|\mathbf{A}\|_F^2 = \sum_{j=1}^{j=n}\sum_{i=1}^{i=r}(\mathbf{a_j}^T\mathbf{v_i})^2 = \sum_{i=1}^{i=r}\|\mathbf{A}\mathbf{v_i}\|^2 = \sum_i {\sigma_i}^2. \tag{268}$$

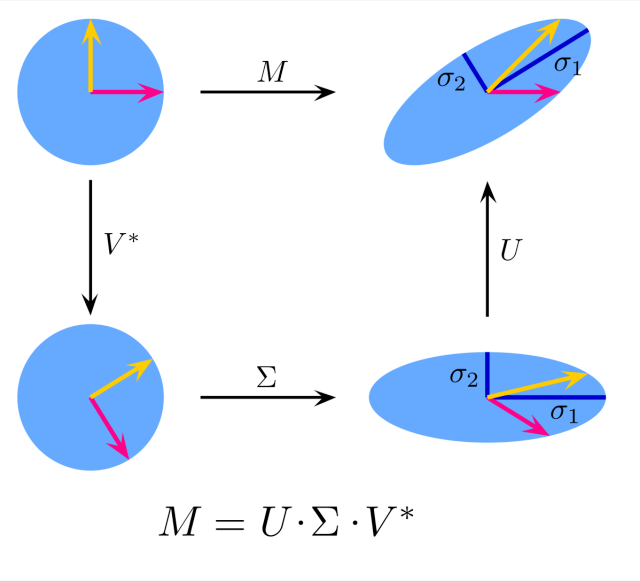


Figure 8: Interpreting SVD

4.8.3 Steps in Transformation

4.8.4 Eigen vectors of AA^T

Let us use the SVD to express the product $\mathbf{AA^T}$.

$$\mathbf{AA^T} = \mathbf{U\Sigma V^T V\Sigma^T U^T} = \mathbf{U\Sigma^2 U^T}. \tag{269}$$

With some reasonable assumptions like A being square and having a decomposition of the form $S\Lambda S^T$ where S is the matrix of eigenvectors as columns, one can see that the eigenvalues of $\mathbf{AA^T}$ are the squares of the singular values and the eigenvector matrix is equivalent to the left singular matrix of $\mathbf{A}$.

5 Calculus

5.1 Numerical Methods for Machine Learning

Calculus is an essential tool for machine learning algorithms, and numerical methods play a critical role in the implementation of these algorithms. In this report, we will discuss the numerical methods used in calculus for machine learning applications. We will start by reviewing the basic concepts of calculus and then move on to the numerical methods used for differentiation and integration. We will also discuss the numerical methods used in optimization problems, such as finding the minimum or maximum value of a function. Finally, we will discuss the use of numerical methods in solving differential equations.

5.2 Revision of Calculus

Calculus is the branch of mathematics that deals with the study of continuous change. It has two main branches: differential calculus and integral calculus. Differential calculus deals with the study of the rate of change of a function, while integral calculus deals with the study of the accumulation of a function over a particular interval.

5.2.1 Derivatives

The derivative of a function is the rate of change of that function with respect to a particular variable. It is a fundamental concept in calculus and is used extensively in machine learning algorithms. The derivative of a function $f(x)$ with respect to x is denoted as $f'(x)$ and is defined as:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

This definition gives us the instantaneous rate of change of a function at a particular point. However, it is not always possible to find the derivative of a function using this definition. In such cases, we use numerical methods to approximate the derivative.

5.2.2 Integrals

The integral of a function is the area under the curve of that function over a particular interval. It is also a fundamental concept in calculus and is used extensively in machine learning algorithms. The integral of a function $f(x)$ over the interval $[a, b]$ is denoted as:

$$\int_a^b f(x)dx$$

The integral is also defined as the antiderivative of the function $f(x)$, denoted as $F(x)$. The fundamental theorem of calculus states that if $F(x)$ is the antiderivative of $f(x)$, then:

$$\int_a^b f(x)dx = F(b) - F(a)$$

However, it is not always possible to find the antiderivative of a function. In such cases, we use numerical methods to approximate the integral.

5.3 Numerical Methods

Numerical methods are used to approximate the derivatives and integrals of a function when it is not possible to find the exact solution. In this section, we will discuss the numerical methods used for differentiation and integration.

5.3.1 Numerical Differentiation

There are several numerical methods used to approximate the derivative of a function. The most common methods are:

- Forward difference method

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

- Backward difference method

$$f'(x) \approx \frac{f(x) - f(x-h)}{h}$$

- Central difference method

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$

Where h is a small positive number called the step size. The smaller the step size, the more accurate the approximation.

5.3.2 Numerical Integration

There are several numerical methods used to approximate the integral of a function. The most common methods are:

- Trapezoidal rule

$$\int_a^b f(x)dx \approx \frac{h}{2}\left[f(a) + 2\sum_{i=1}^{n-1} f(x_i) + f(b)\right]$$

  where $h = \frac{b-a}{n}$ and $x_i = a + ih$.

- Simpson's rule

$$\int_a^b f(x)dx \approx \frac{h}{3}\left[f(a) + 4\sum_{i=1}^{n/2} f(x_{2i-1}) + 2\sum_{i=1}^{n/2-1} f(x_{2i}) + f(b)\right]$$

where $h = \frac{b-a}{n}$ and $x_i = a + ih$.

5.4 Optimization

Optimization problems involve finding the minimum or maximum value of a function. These problems are common in machine learning algorithms. There are several numerical methods used to solve optimization problems. The most common methods are:

- Gradient descent

Gradient descent is an iterative optimization algorithm that uses the gradient of the function to find the minimum or maximum value. It starts with an initial guess and updates the guess iteratively until it converges to the minimum or maximum value.

Gradient Descent Algorithm

Data: Initialise θ, α
while $\|\theta^{(i)} - \theta^{(i-1)}\| > \epsilon$ **do**
  $\theta \leftarrow \theta - \alpha\nabla J(\theta)$ `// update parameters`
end

In this algorithm, θ represents the vector of parameters that we want to optimize,
$J(\theta)$ is the cost function we want to minimize, and
$\nabla J(\theta)$ is the gradient of the cost function with respect to the parameters.
α is the learning rate and determines the step size we take at each iteration.

5.5 Conclusion

Numerical methods play a critical role in calculus for machine learning algorithms. In this report, we have discussed the numerical methods used for differentiation, integration and optimization. These methods are essential tools for solving mathematical problems that arise in machine learning. By using numerical methods, we can approximate complex functions, solve optimization problems, and model dynamic systems.
It is important to note that numerical methods are not always exact and can introduce errors into our calculations. The accuracy of the results depends on the step size, the choice of method, and the properties of the function being approximated. Therefore, it is important to carefully choose the appropriate method and step size for a given problem to ensure accurate results.
In conclusion, numerical methods are fundamental to calculus in machine learning. They allow us to approximate functions, solve optimization problems, and model dynamic systems. By understanding and applying numerical methods effectively, we can develop accurate and efficient machine learning algorithms.

5.6 Reference

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